Gestion des projets IA

Méthode bout en bout

Objectif

Vous faire découvrir les différentes méthodes auxquelles vous serez confrontés dans la gestion de projet d’IA

Limites

C’est quelque chose qui dure 200h/6 mois pour 9000€.

https://imt-nord-europe.fr/annuaire-formations/management-de-projets-en-intelligence-artificielle/

Version OpenClassrooms sur 3 mois …

Materials

- Agile
- Cycle en V

Méthode ?

Atelier

Use this slide to show a major stat. It can help enforce the presentation’s main message or argument.

- Distance
- Spécficiations
- Surstock
- Redondance
- Delais
- *Material*
- …

Atelier

Use this slide to show a major stat. It can help enforce the presentation’s main message or argument.

Les workflows

Dominos Pizza 🍕 App & JIRA Board Workflows. What’s the connection?

Méthode Scrum : présentation et avantages

Kanban

a scheduling system for lean manufacturing to improve manufacturing efficiency. Taiichi Ohno, an industrial engineer at Toyota,

takes its name from the cards that track production within a factory.

problem areas are highlighted by measuring lead time and cycle time of the full process and process steps.

One of the main benefits of kanban is to establish an upper limit to work in process (commonly referred as "WIP") inventory to avoid overcapacity.

Kanban

To limit the buildup of excess inventory at any point in production.

Limits on the number of items waiting at supply points are established and then reduced as inefficiencies are identified and removed.

Whenever a limit is exceeded, this points to an inefficiency that should be addressed.



Kanban (cards)

key component : they **signal the need to move materials within a production** facility or to move materials from an outside supplier into the production facility.

is a message that signals a **depletion of product, parts, or inventory**. When received, the kanban triggers replenishment of that product, part, or inventory. Consumption, therefore, drives demand for more production, and the kanban card signals demand for more product.

so kanban cards help create a **demand-driven system**, leading to faster turnarounds in production and lower inventory levels, helping companies implementing such systems be more competitive.

Scrum

présentée pour la première fois en 1986 dans un article rédigé par Hirotaka Takeuchi et Ikujiro Nonaka pour la Harvard Business Review,

« Scrum » : « au rugby, le ballon passe de main en main dans l'équipe, tandis que cette dernière se déplace sur le terrain comme un seul homme. »

approfondie et codifiée par Ken Schwaber et Jeff Sutherland en 1995 dans leurs publications Agile Manifesto et SCRUM Development Process.

Scrum

a **Product Owner** responsible for detailing the business requirements and ensuring that the business gets a good return on investment.

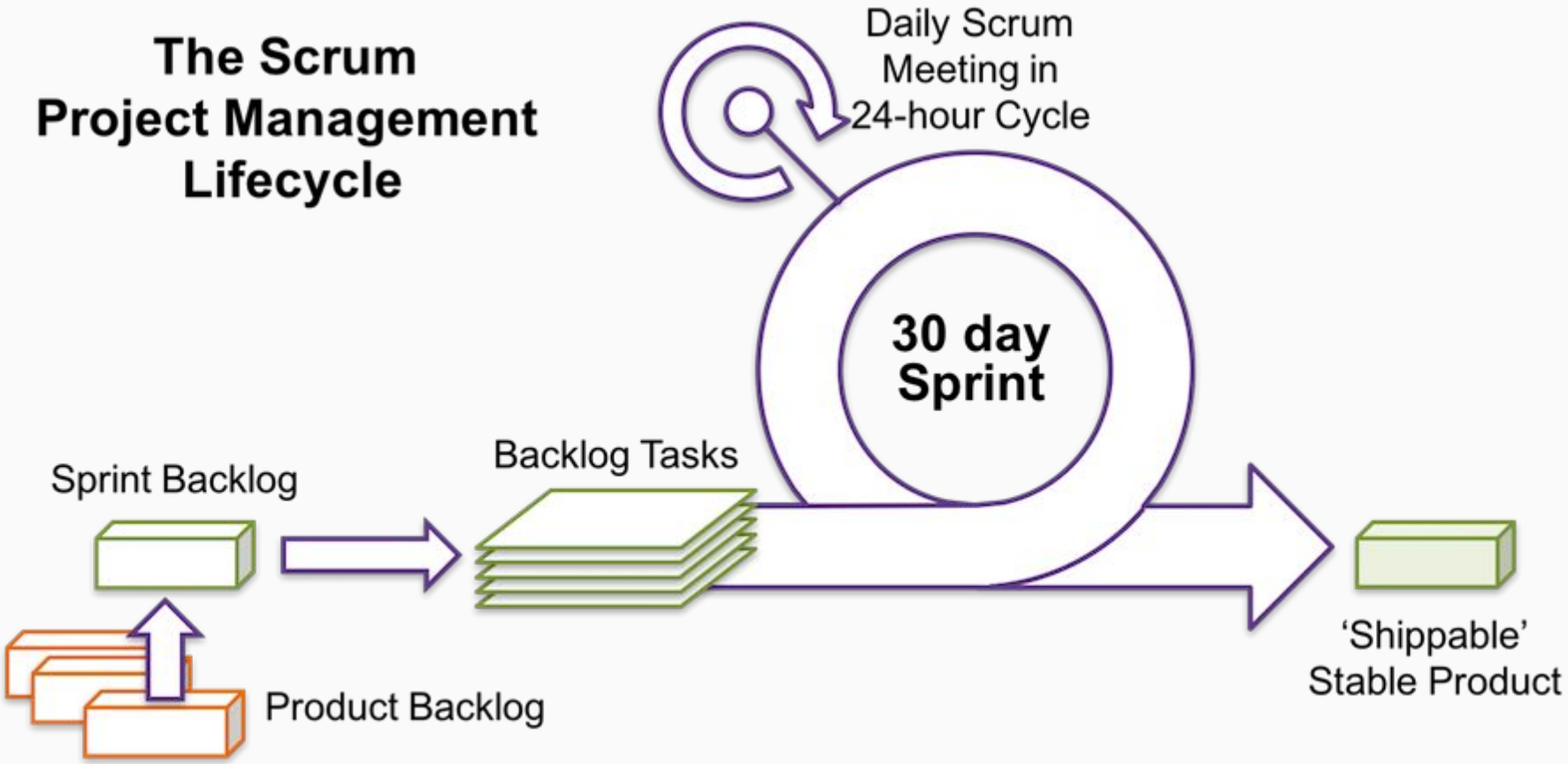
The Scrum Team, led by a **Scrum Master**, selects one subset of functionality from a product backlog of undeveloped functions, divides it into tasks, and works intensively on developing the outputs for a fixed time: Sprint (~ 30 days)

daily Scrum Meeting to share learning, report progress, discuss challenges, and solve problems.

At the **end of the sprint**, the team should produce **a working product** that is stable and useful.

After a reflection and learning process, the team then works with the product owner to define the subset of functionality it will work on in the **next sprint**.

Kanban (cards)

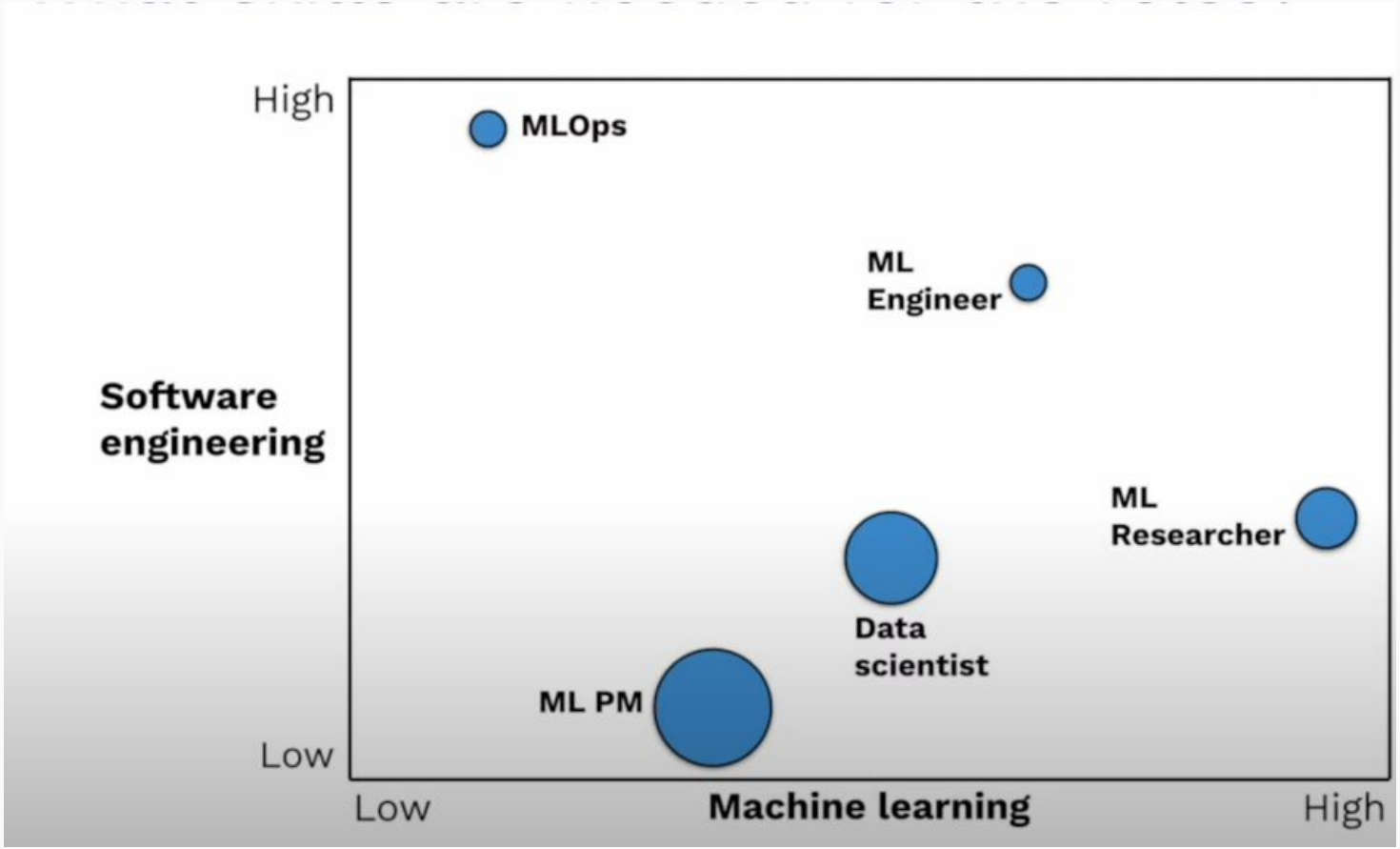


Et en IA

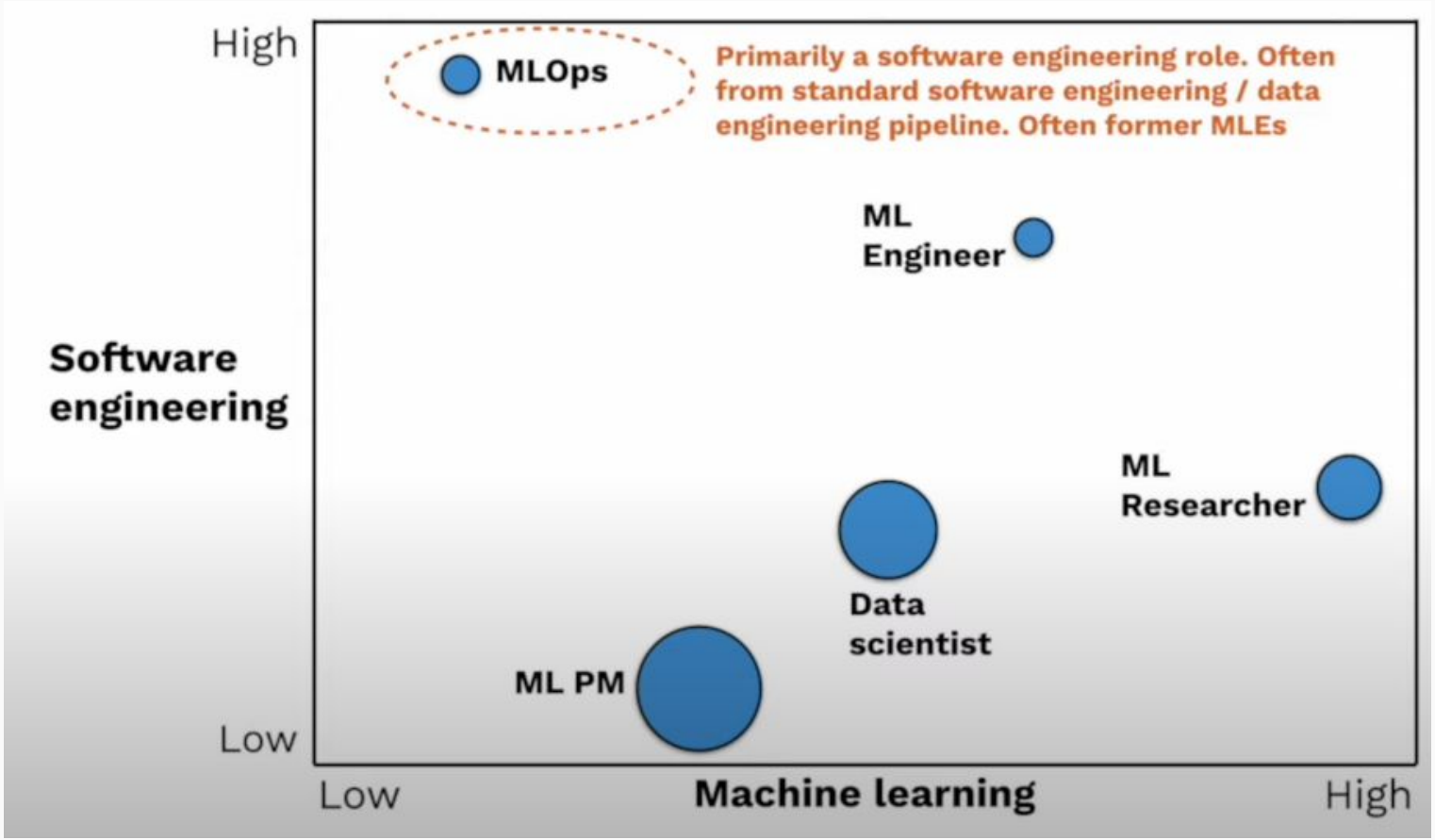
Ressources

Role	Job Function	Work product
ML product manager	Work with ML team, business, users, data owners to prioritize & execute projects	Design docs, wireframes, work plans
MLOps / ML platform	Build the infrastructure to make models easier to deploy, more scalable, etc	ML infrastructure
ML engineer	Train, deploy, & maintain prediction models	Prediction system running on real data in production
ML researcher	Train prediction models (often forward looking or not production-critical)	Prediction model & report describing it
Data scientist	Blanket term used to describe all of the above. In some orgs, means answering business questions using analytics	Prediction model or report

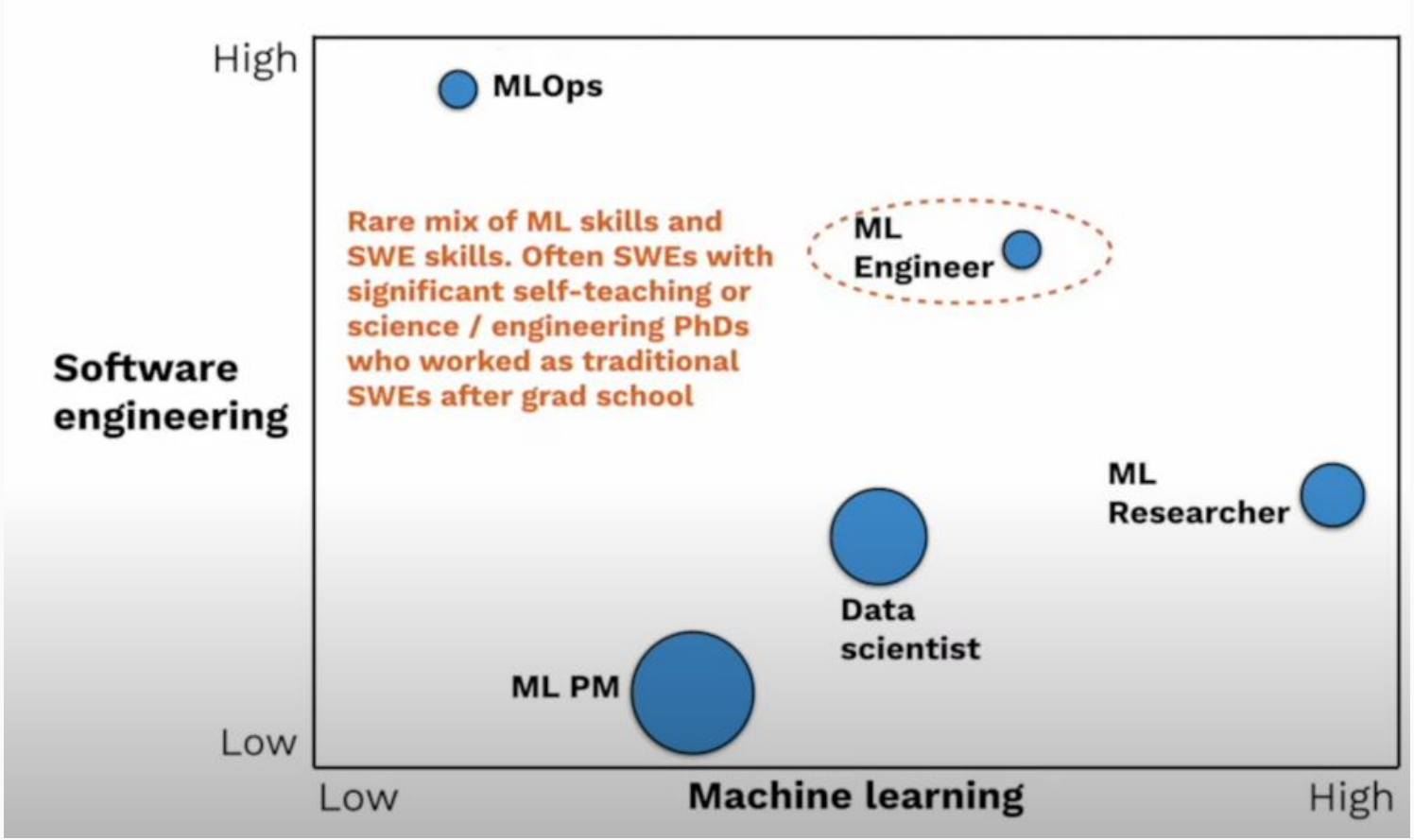
Ressources


High
MLOps
ML
Engineer
Software
engineering
ML
Researcher
Data
scientist
ML PM
Low
Low
Machine learning
High

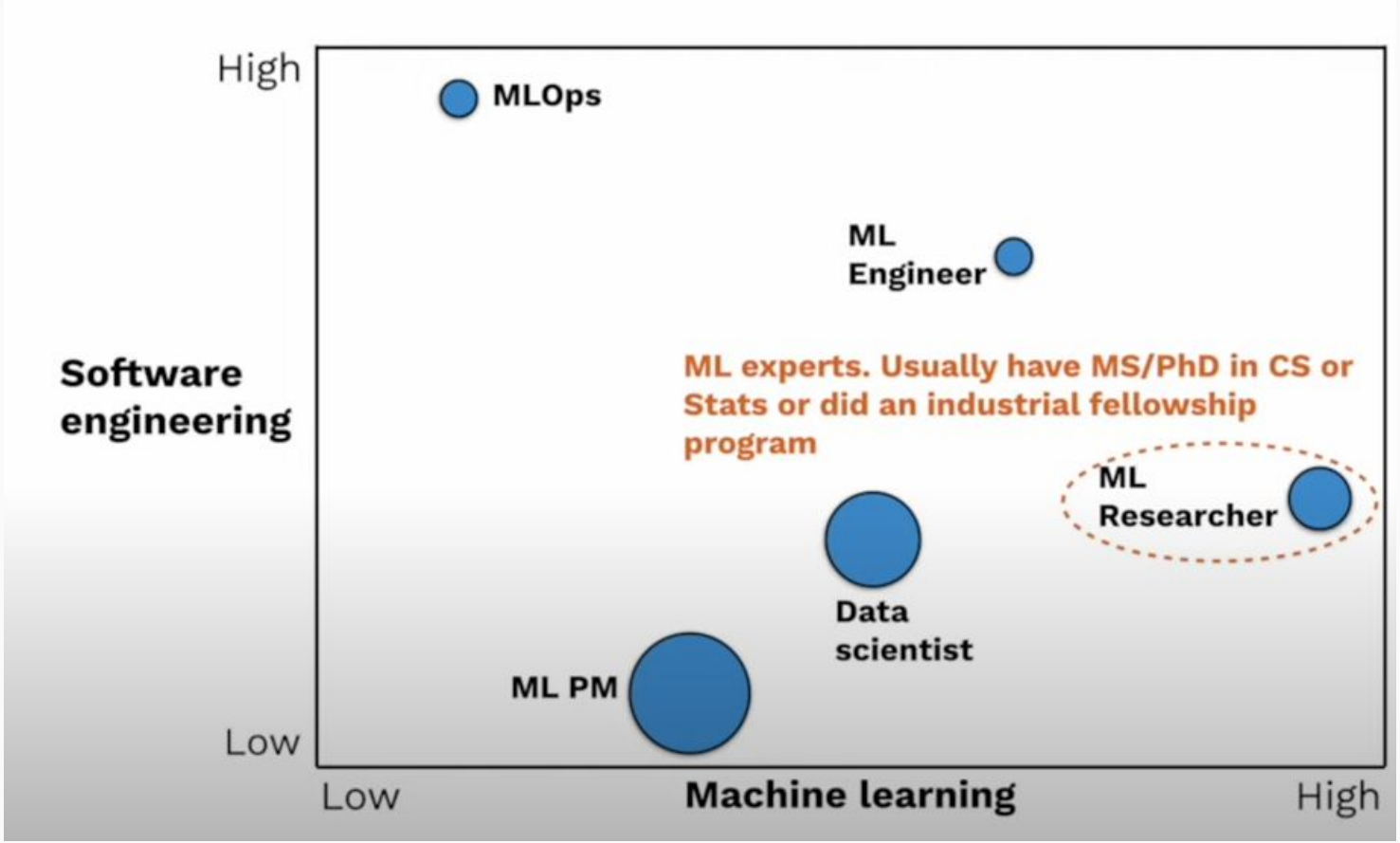
Ressources


High
MLOps
Primarily a software engineering role. Often from standard software engineering / data engineering pipeline. Often former MLEs
ML Engineer
Software engineering
ML Researcher
Data scientist
ML PM
Low
Low
Machine learning
High

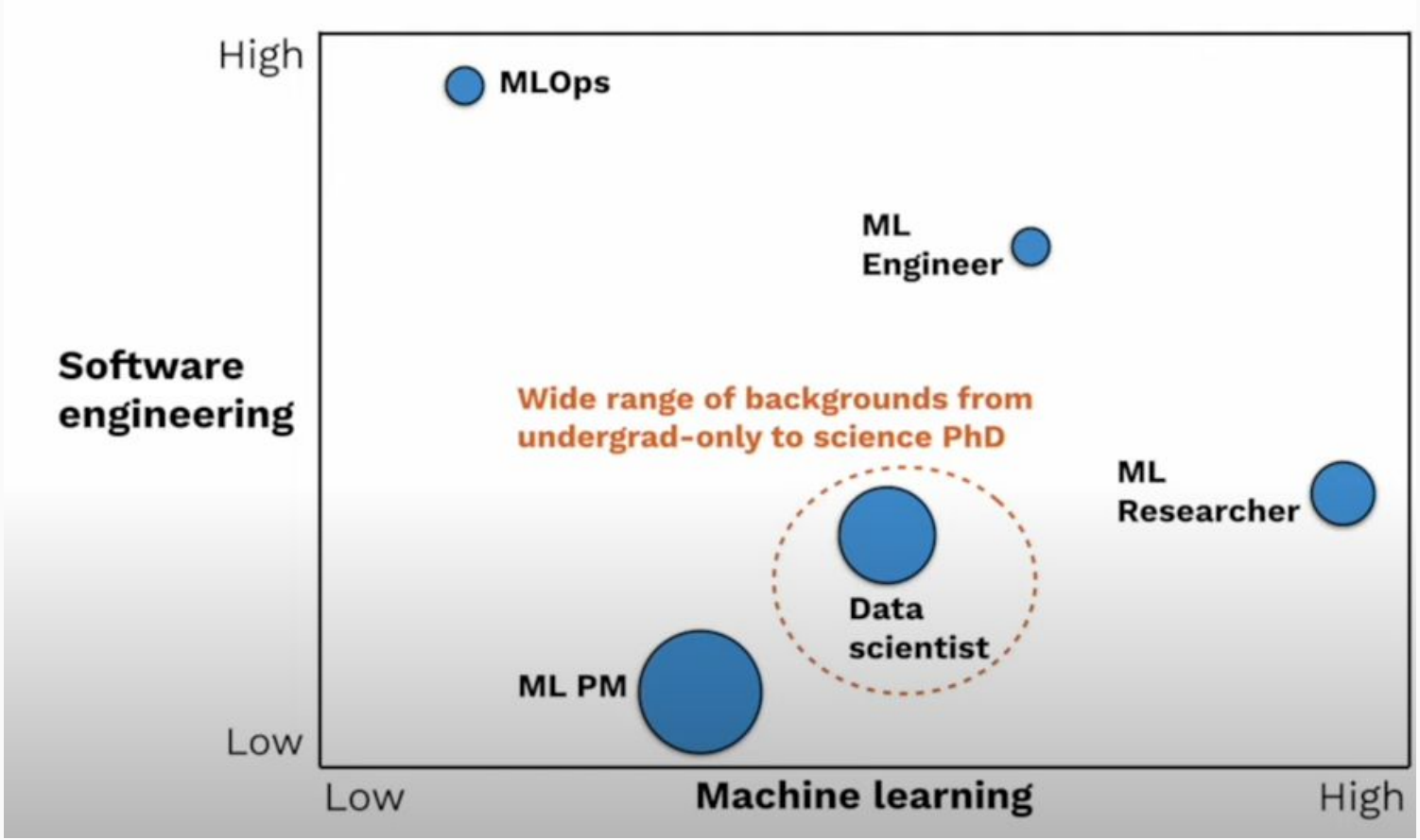
Ressources


High
MLOps
Rare mix of ML skills and SWE skills. Often SWEs with significant self-teaching or science / engineering PhDs who worked as traditional SWEs after grad school
ML Engineer
Software engineering
ML Researcher
Data scientist
ML PM
Low
Low
Machine learning
High

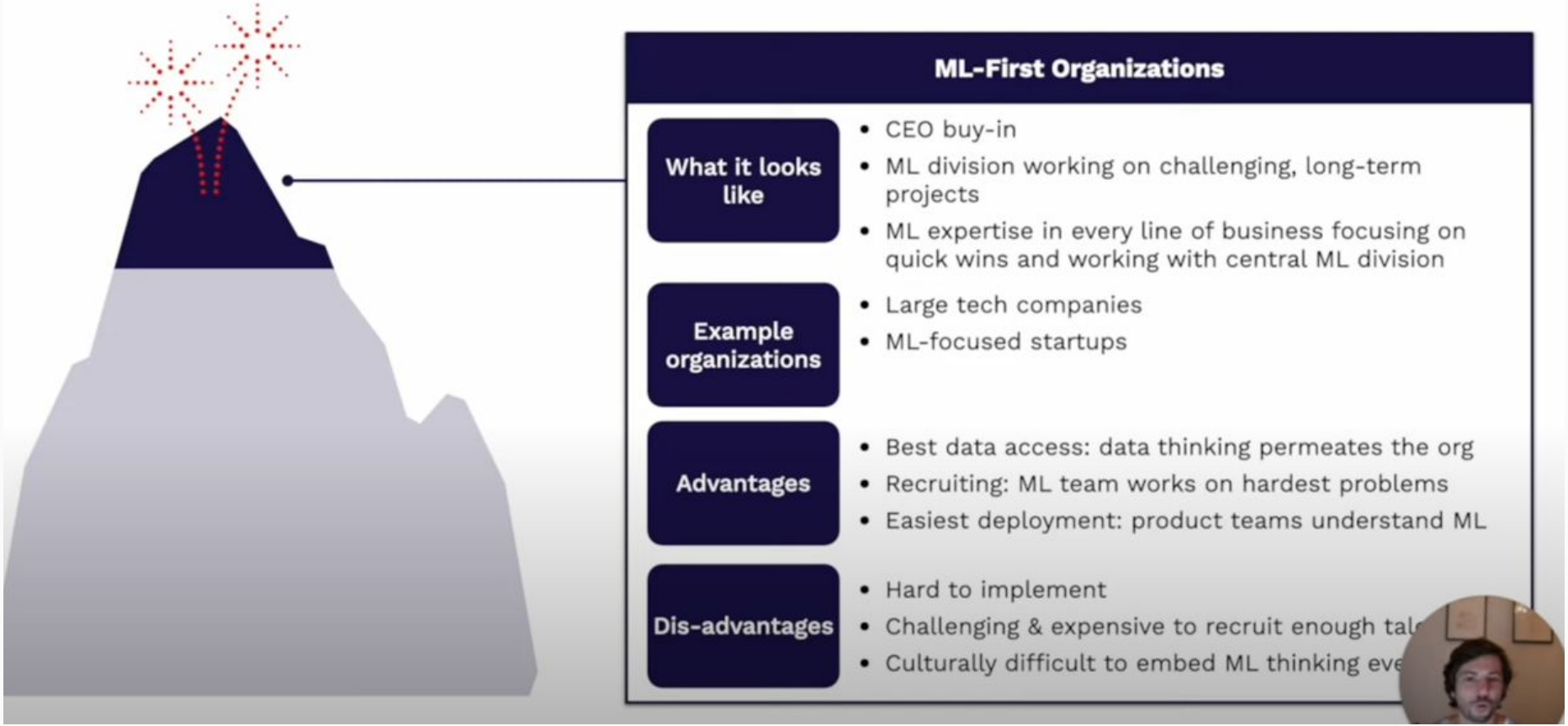
Ressources


High
MLOps
ML
Engineer
Software
engineering
ML experts. Usually have MS/PhD in CS or
Stats or did an industrial fellowship
program
ML
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Data
scientist
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Low
Low
Machine learning
High

Ressources

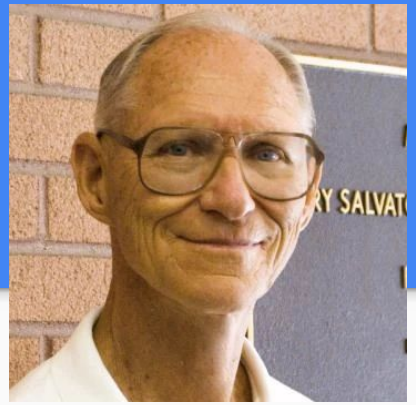

High
MLOps
ML
Engineer
Software
engineering
Wide range of backgrounds from
undergrad-only to science PhD
ML
Researcher
Data
scientist
ML PM
Low
Low
Machine learning
High

Ressources


ML-First Organizations
What it looks like
CEO buy-in
ML division working on challenging, long-term projects
ML expertise in every line of business focusing on quick wins and working with central ML division
Example organizations
Large tech companies
ML-focused startups
Advantages
Best data access: data thinking permeates the org
Recruiting: ML team works on hardest problems
Easiest deployment: product teams understand ML
Dis-advantages
Hard to implement
Challenging & expensive to recruit enough tal
Culturally difficult to embed ML thinking eve

Cycle en W

Le cycle en V



Barry W. Boehm, Software engineering-as it is, 1979.

Issu du monde de l'industrie, le cycle en V est devenu un standard de fait dans l'industrie du logiciel depuis les années 1980. Avec l'apparition de l'ingénierie des systèmes, c'est devenu un standard conceptuel dans de nombreux domaines de l'industrie.

Une alternative **plus réaliste que le modèle en cascade**, en raison de sa prise en compte d'étapes supplémentaires lors de la validation pour distinguer les tests unitaires, les tests d'intégration et les tests systèmes, ce qui s'avère plus adapté aux systèmes complexes faits de plusieurs composants.

End-to-end

Développée par le consortium Confiance.AI

Cadre de gestion de projet pour le développement de systèmes d'IA.

Garantit la qualité et la fiabilité des systèmes d'IA.

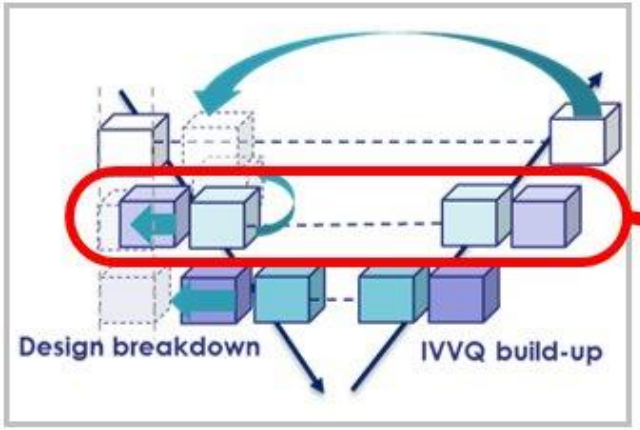
Adapté aux projets d'IA complexes.

conformes à ces normes:

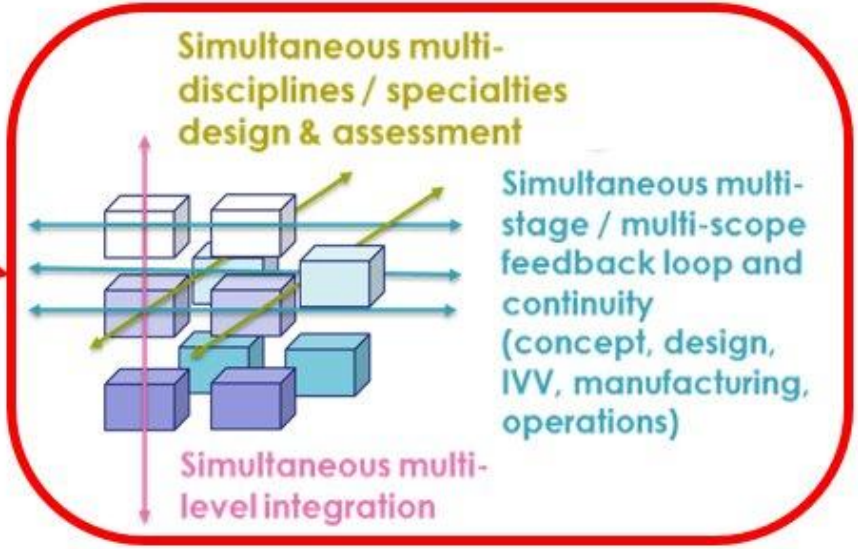
- ISO 5338 Generic AI
- IEEE 7000: éthique.
- SAE ARP 6983 standard sécurité aéronautique.

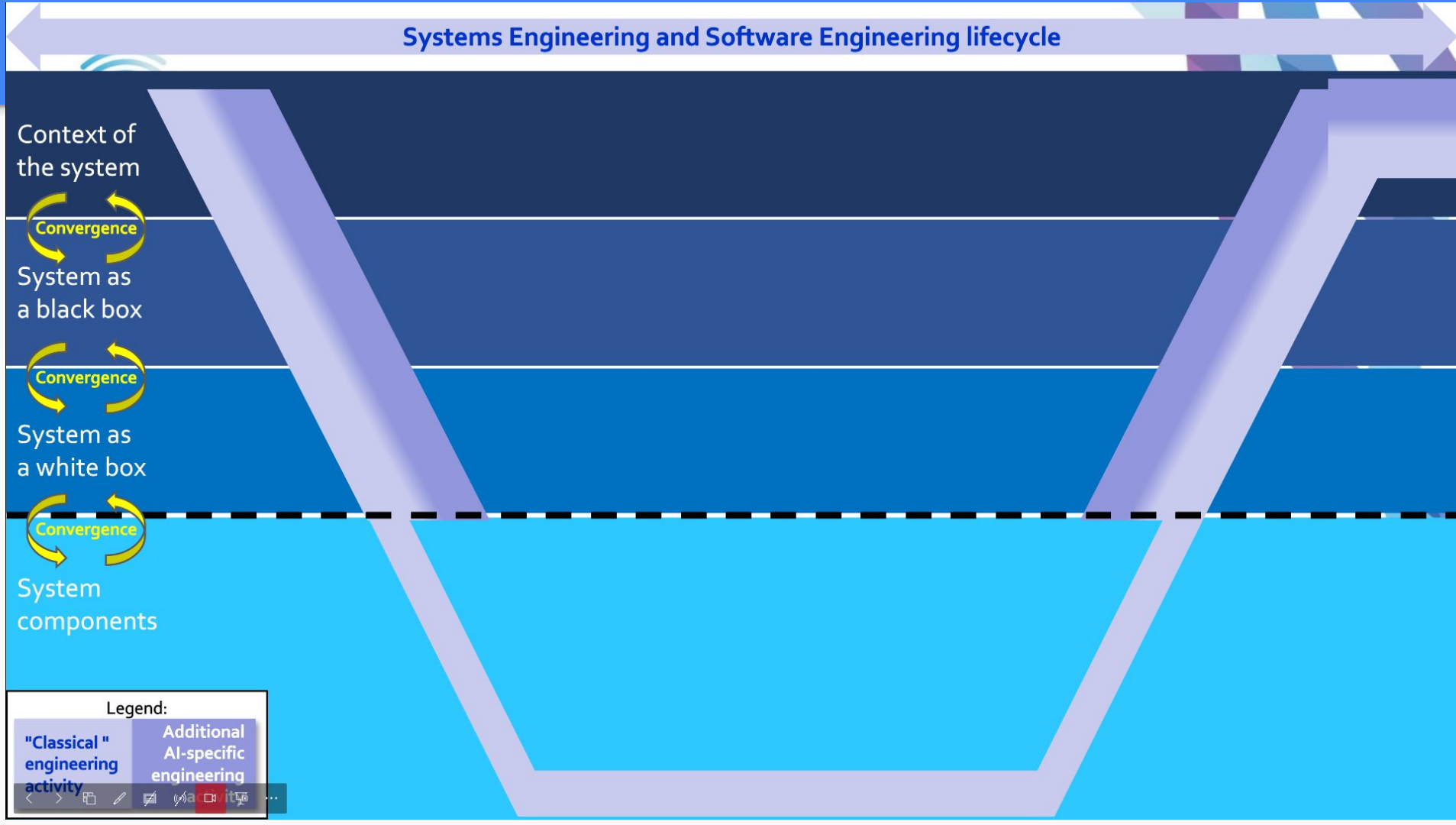


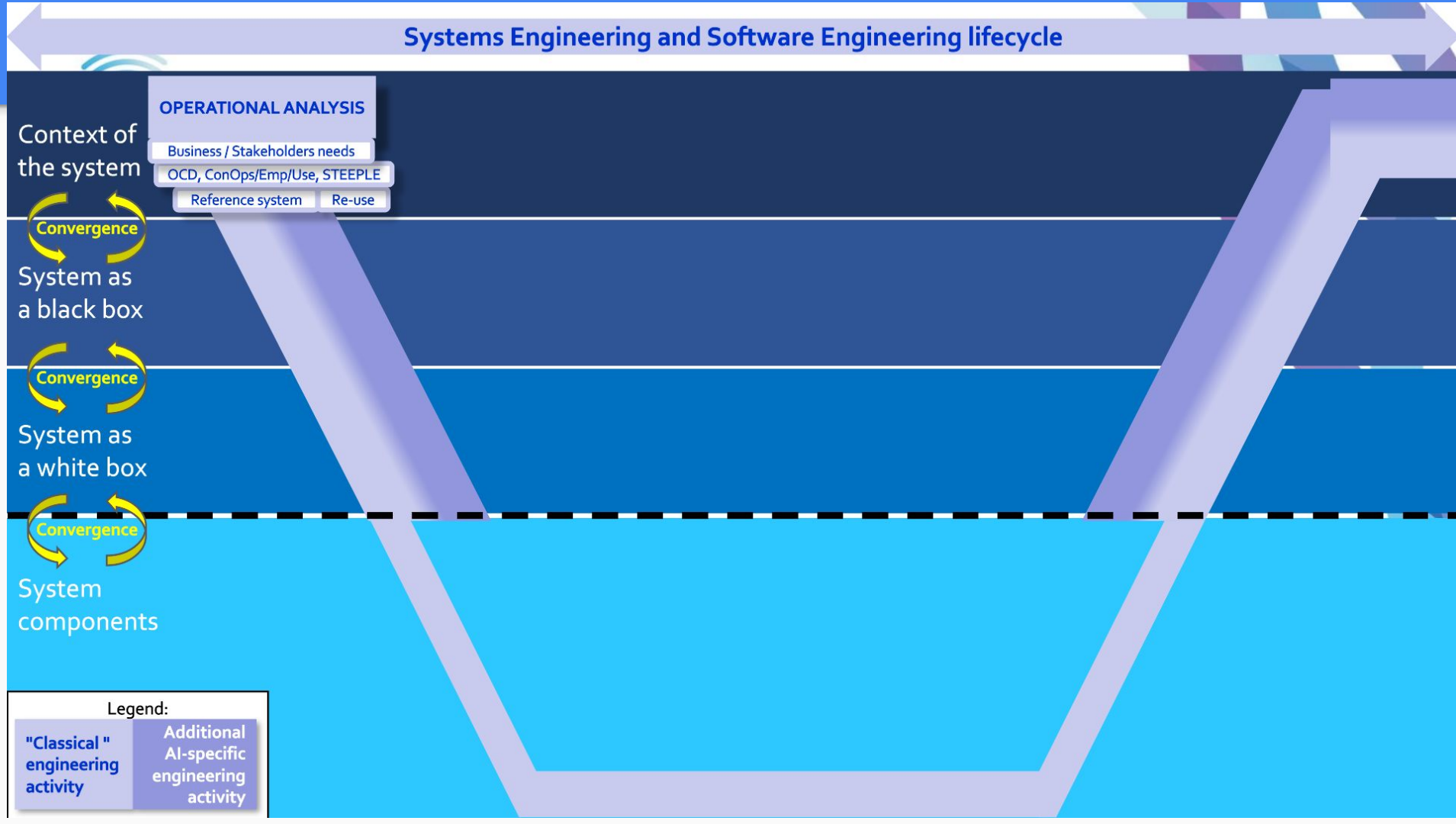
End-to-end: complexity

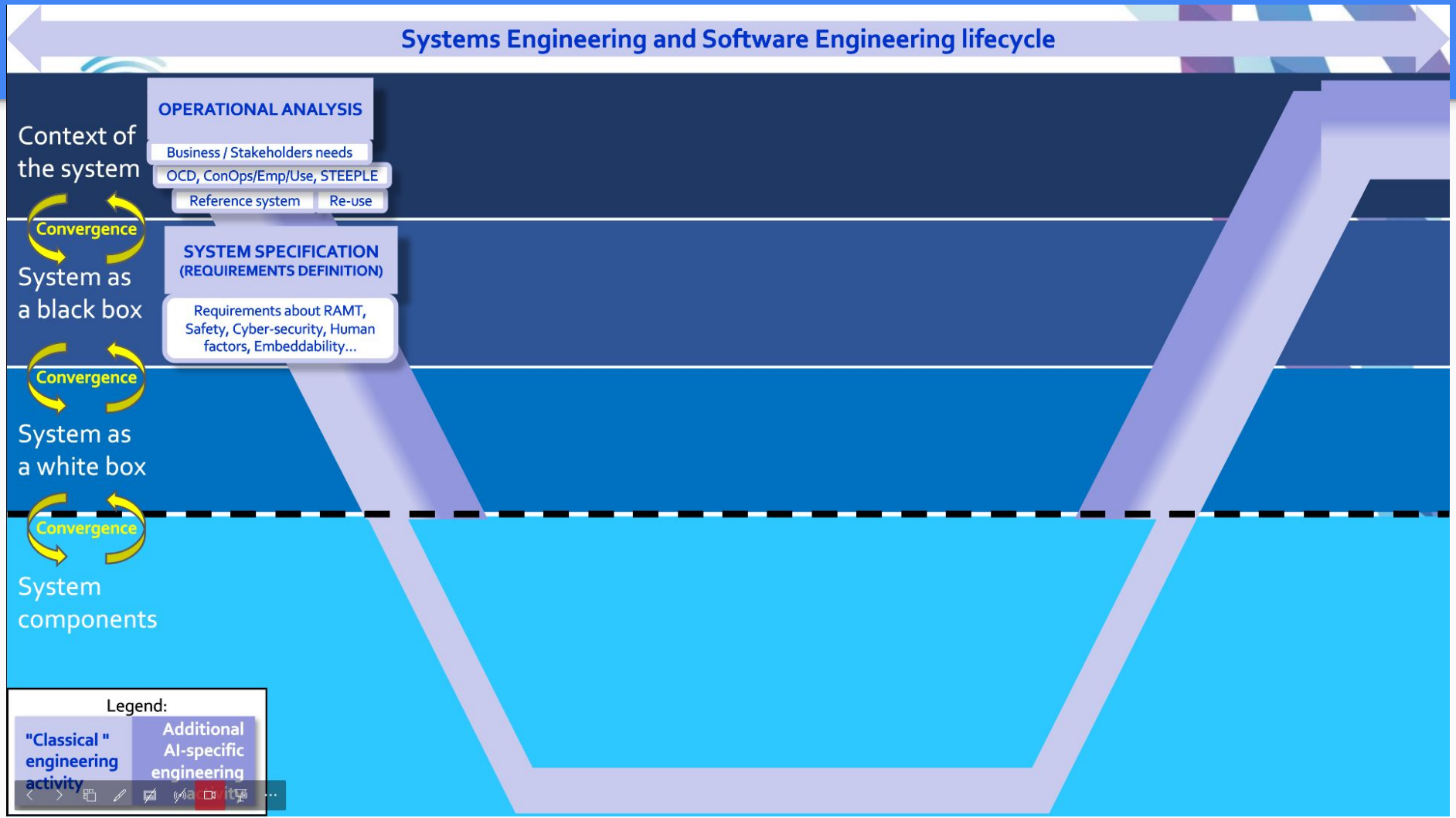


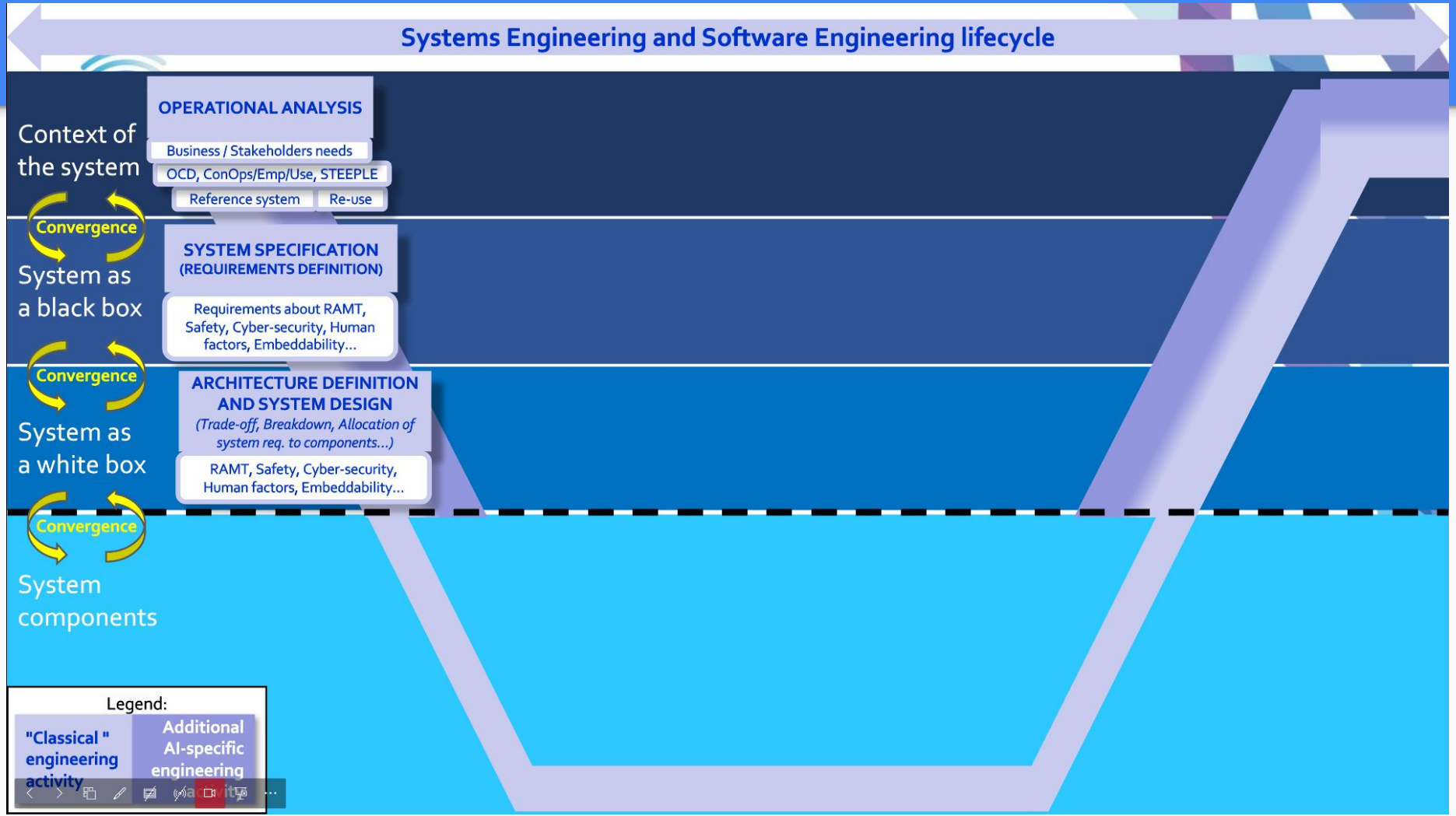


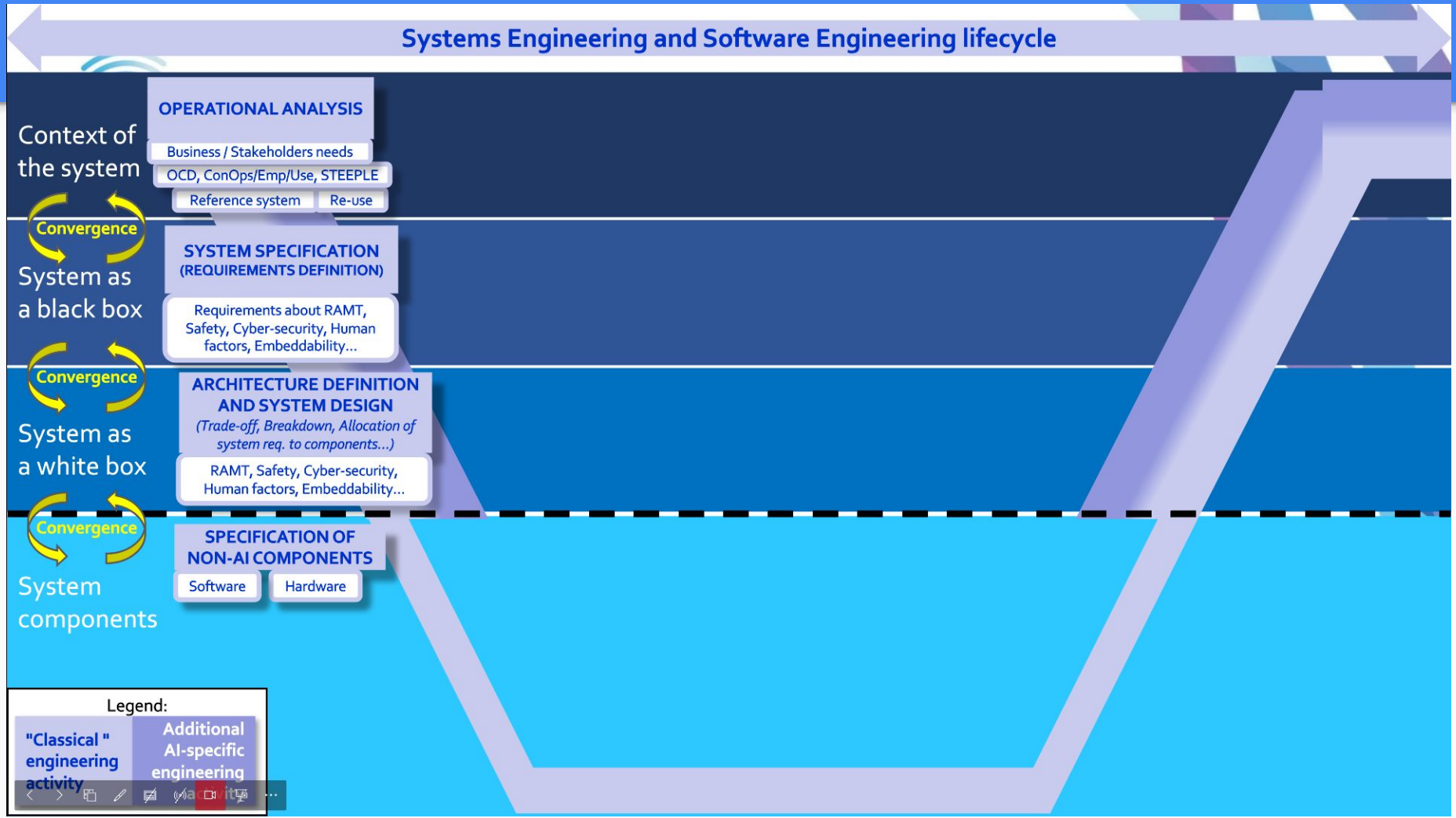


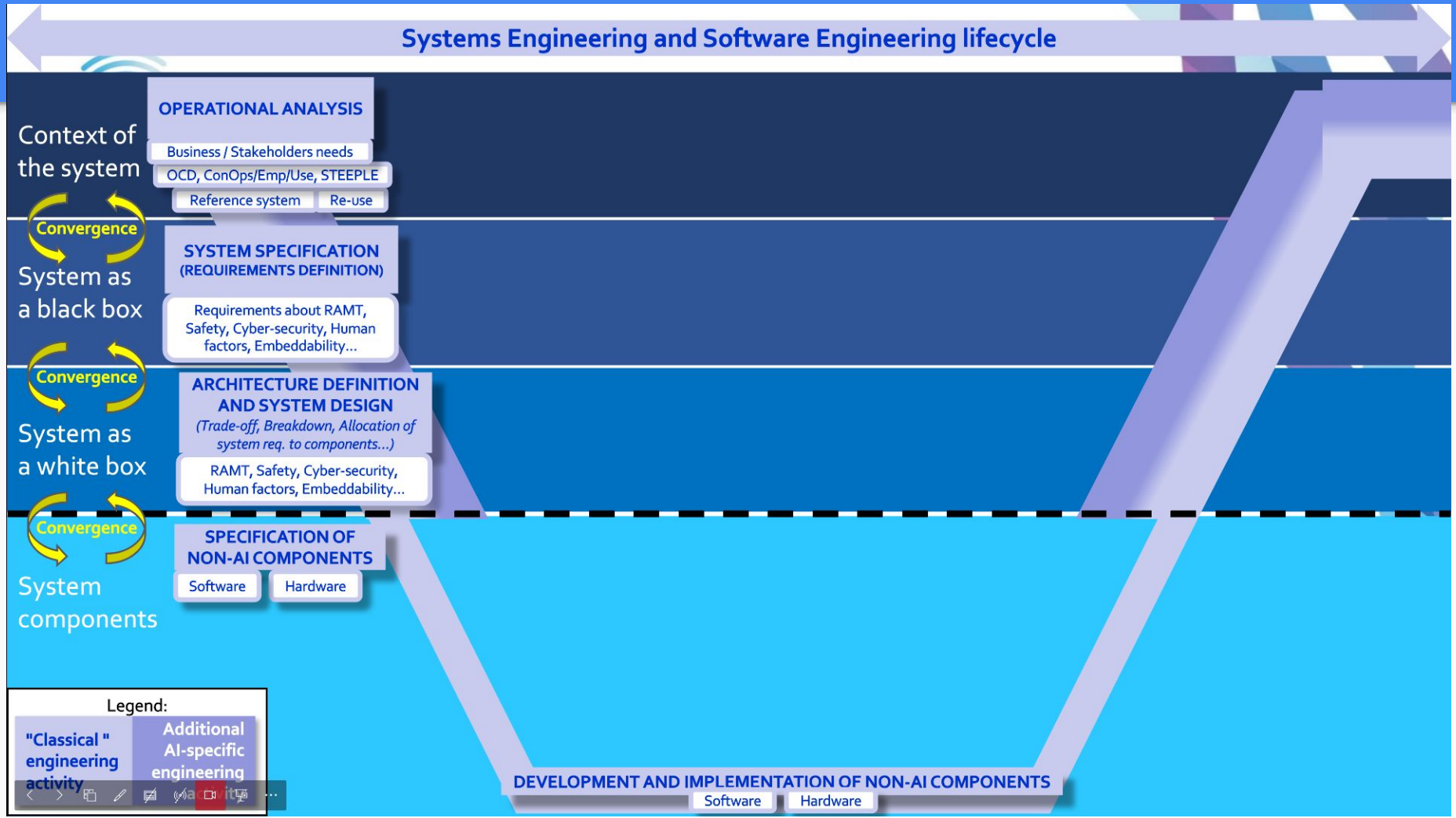

Systems Engineering and Software Engineering lifecycle
Context of the system
Convergence
System as a black box
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System as a white box
Convergence
System components
Legend:
"Classical " engineering activity
Additional AI-specific engineering

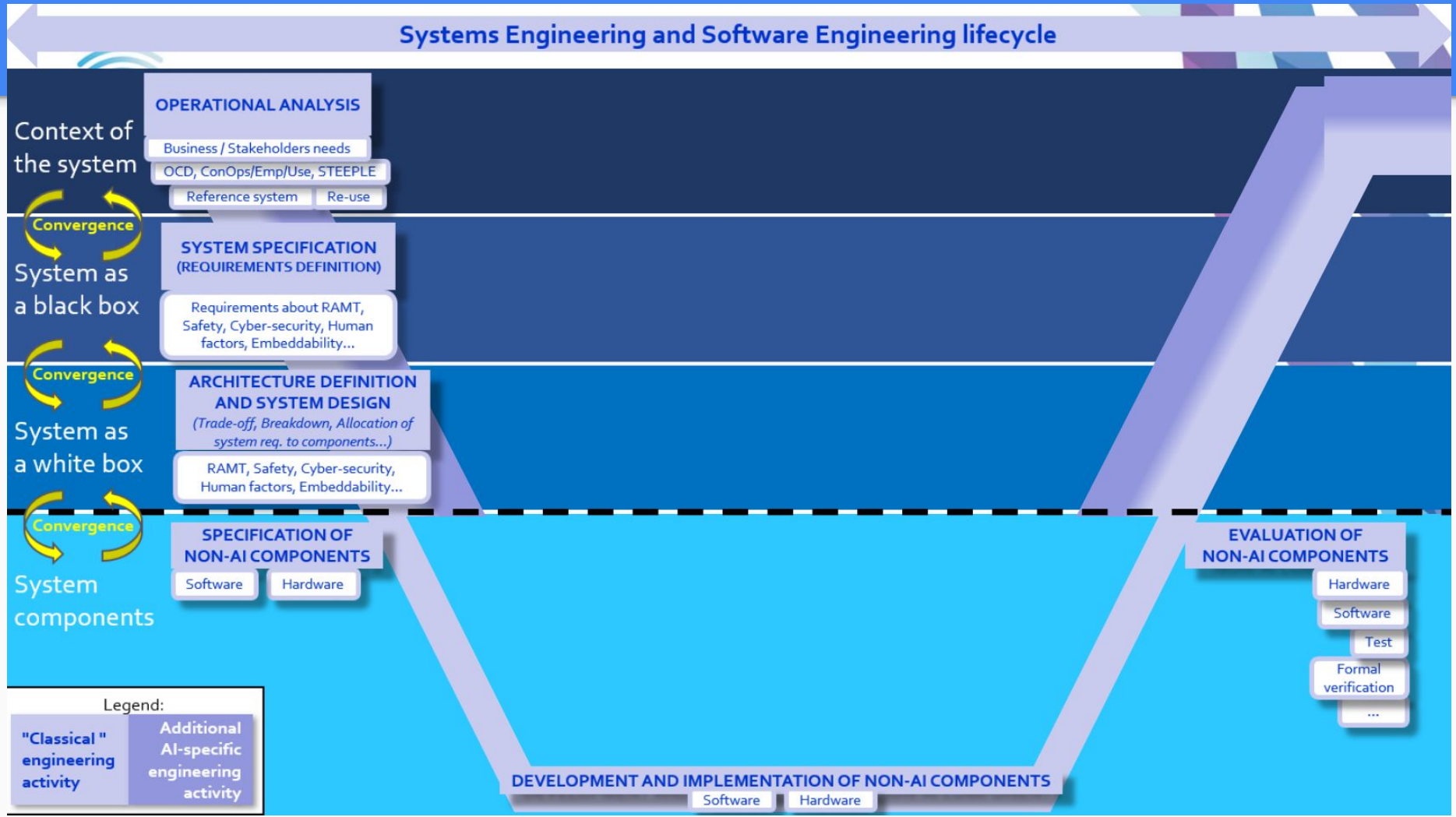

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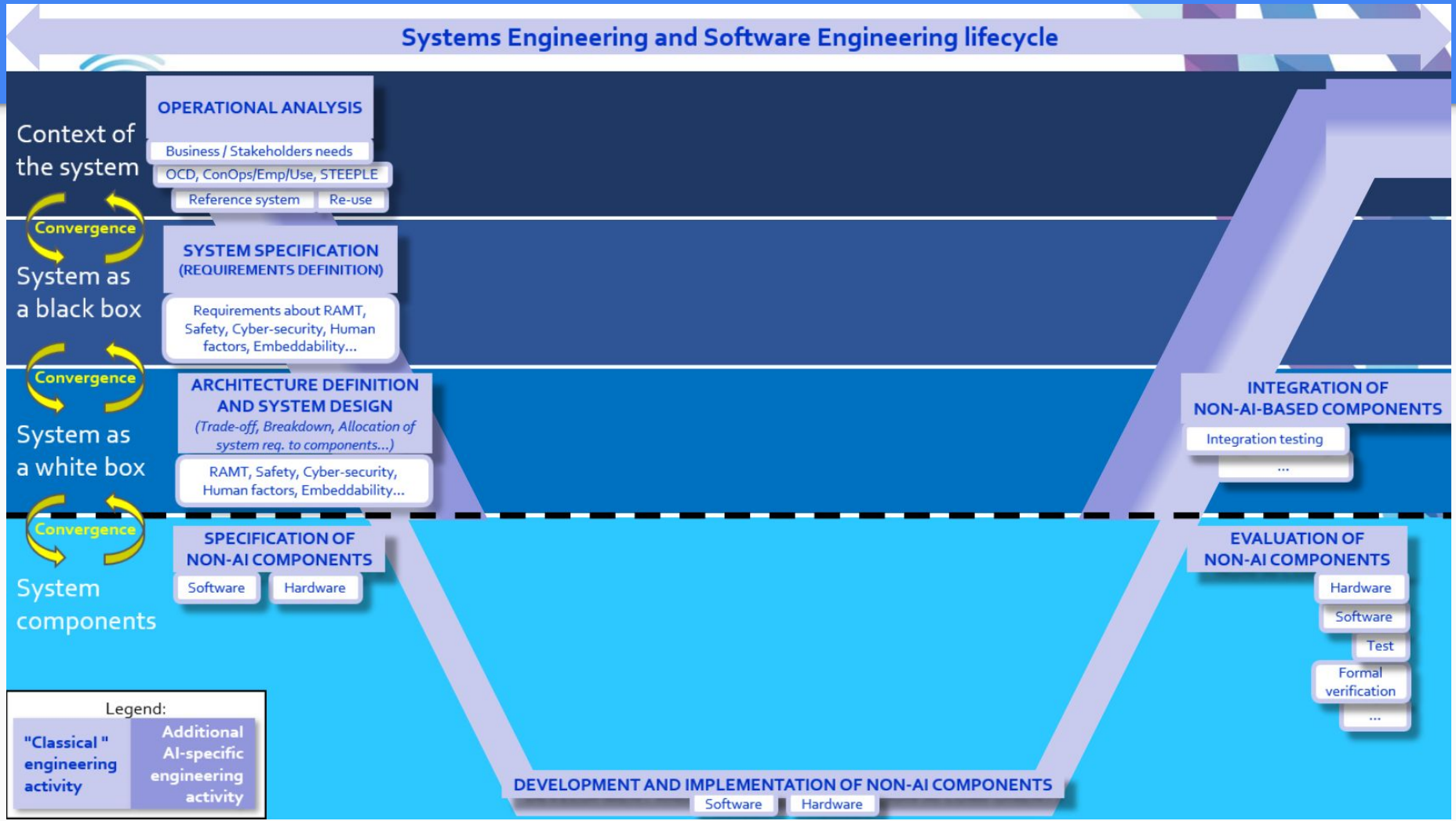

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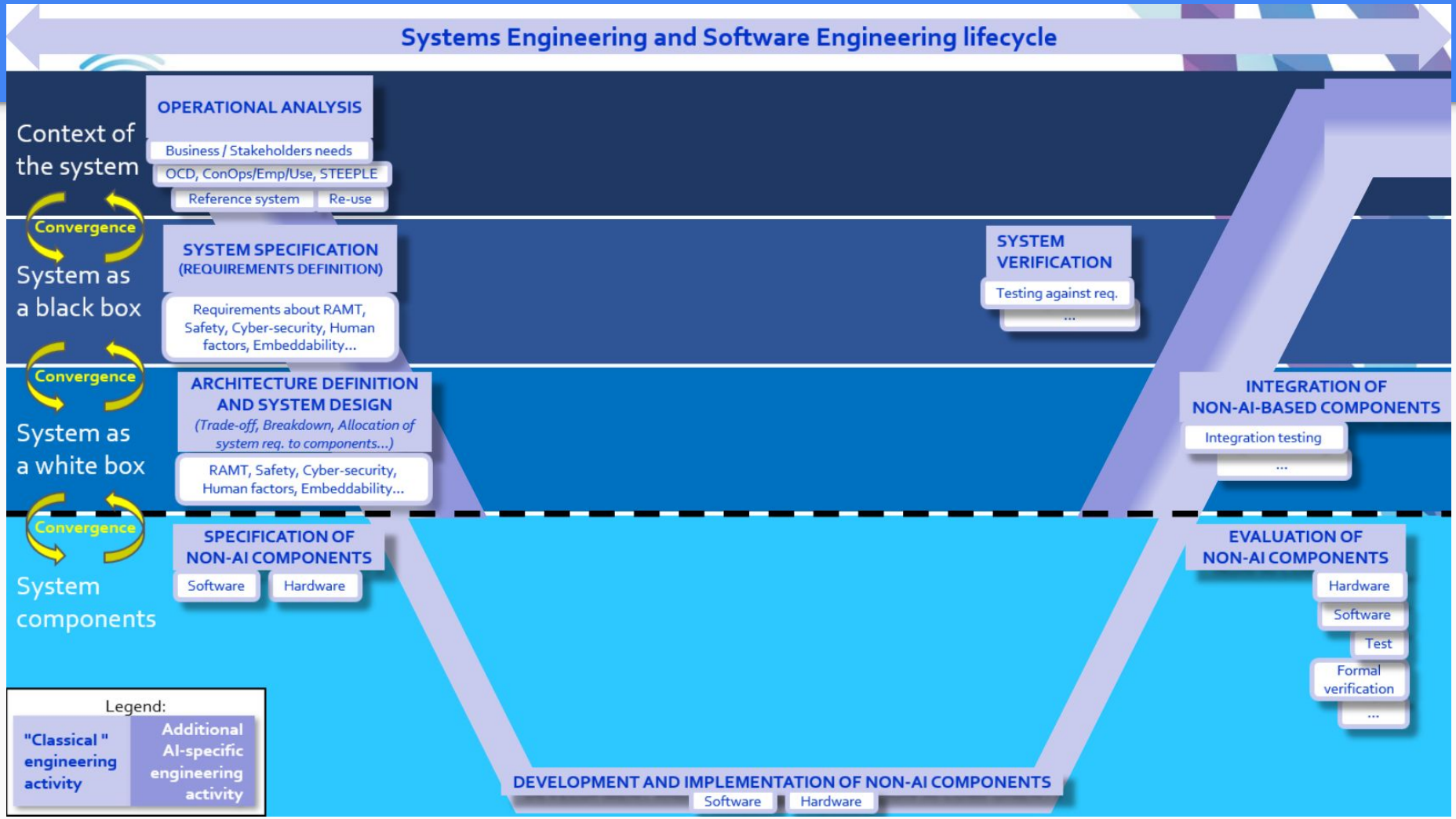

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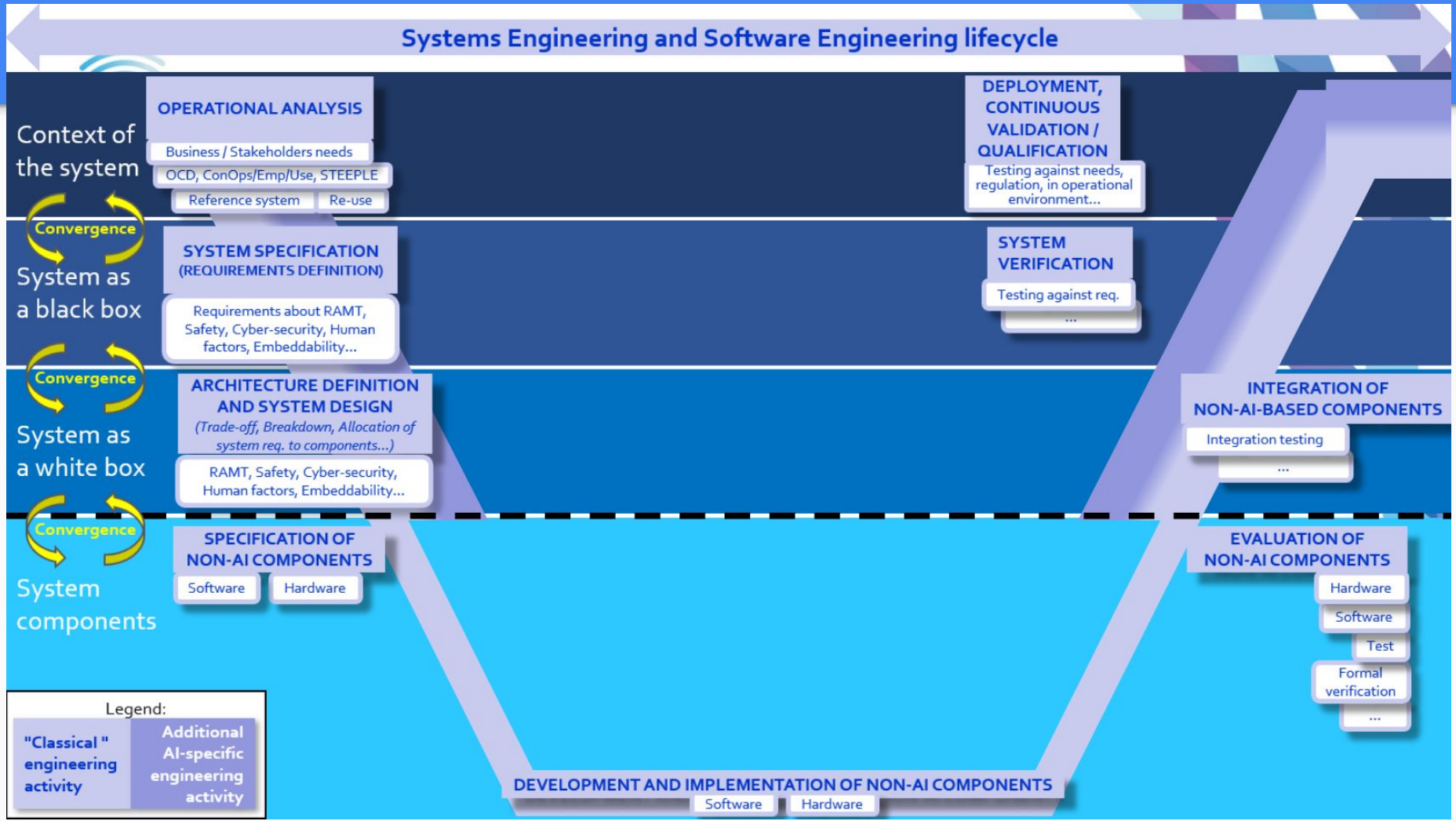

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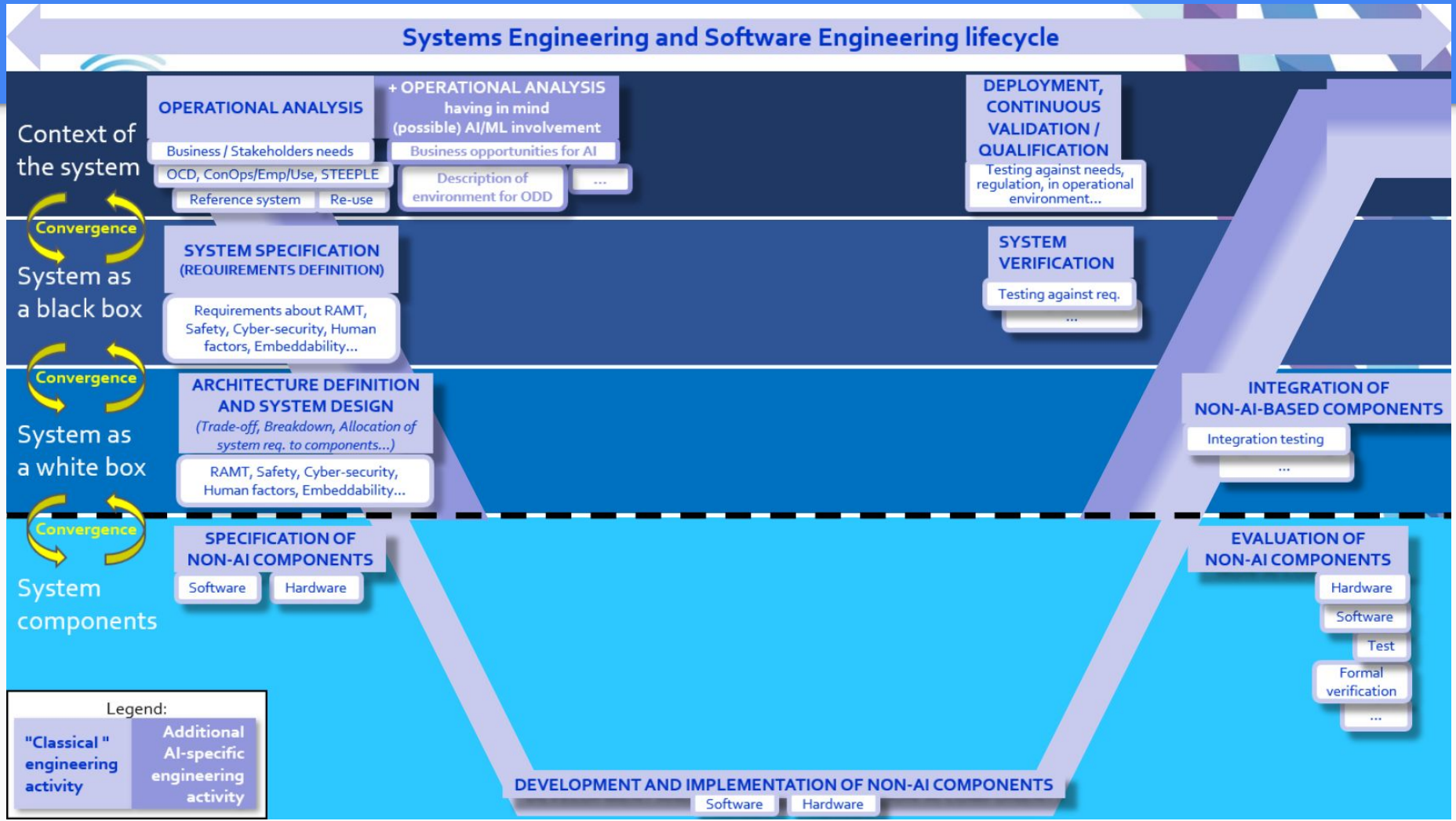

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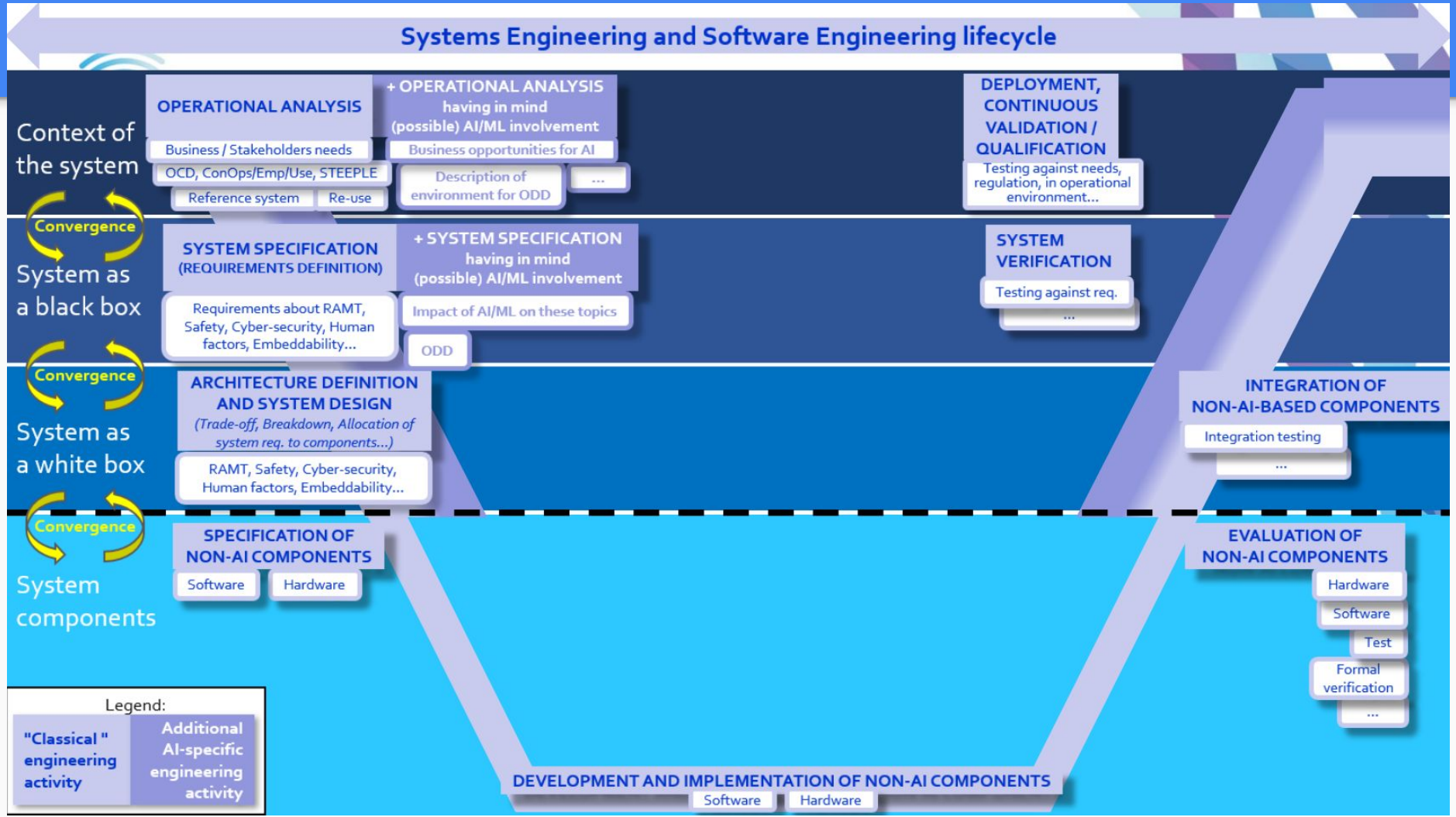

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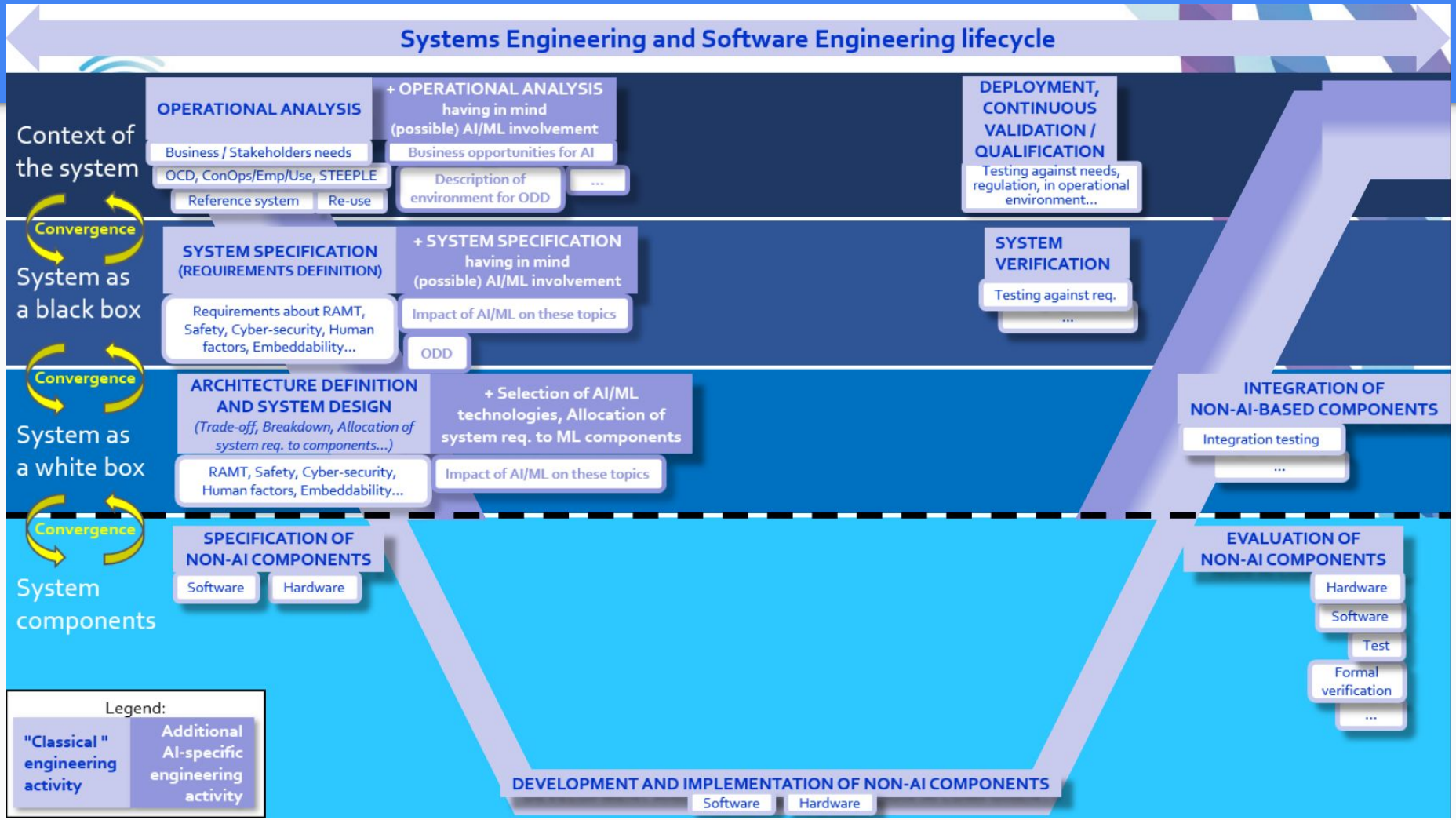

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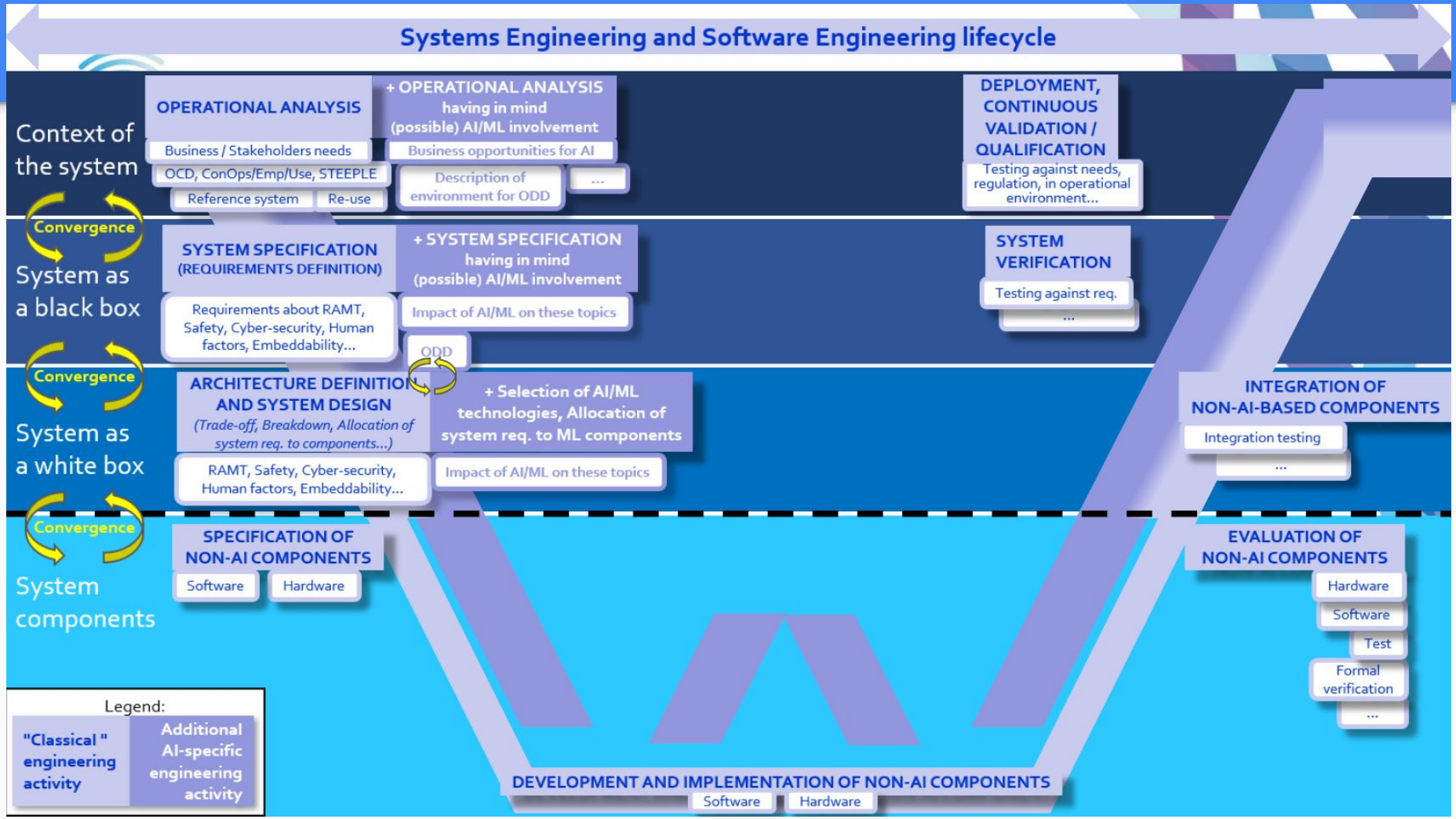

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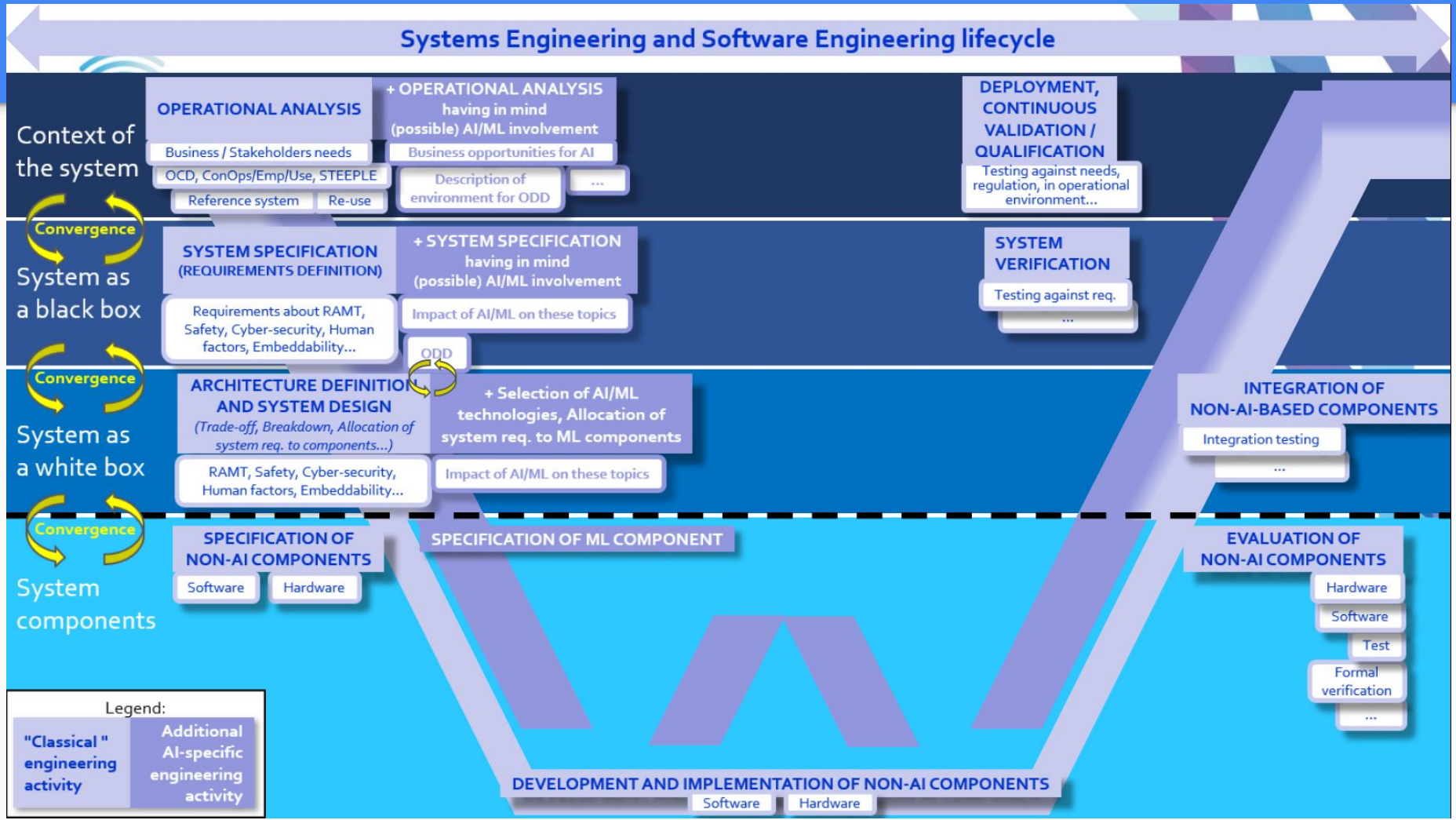

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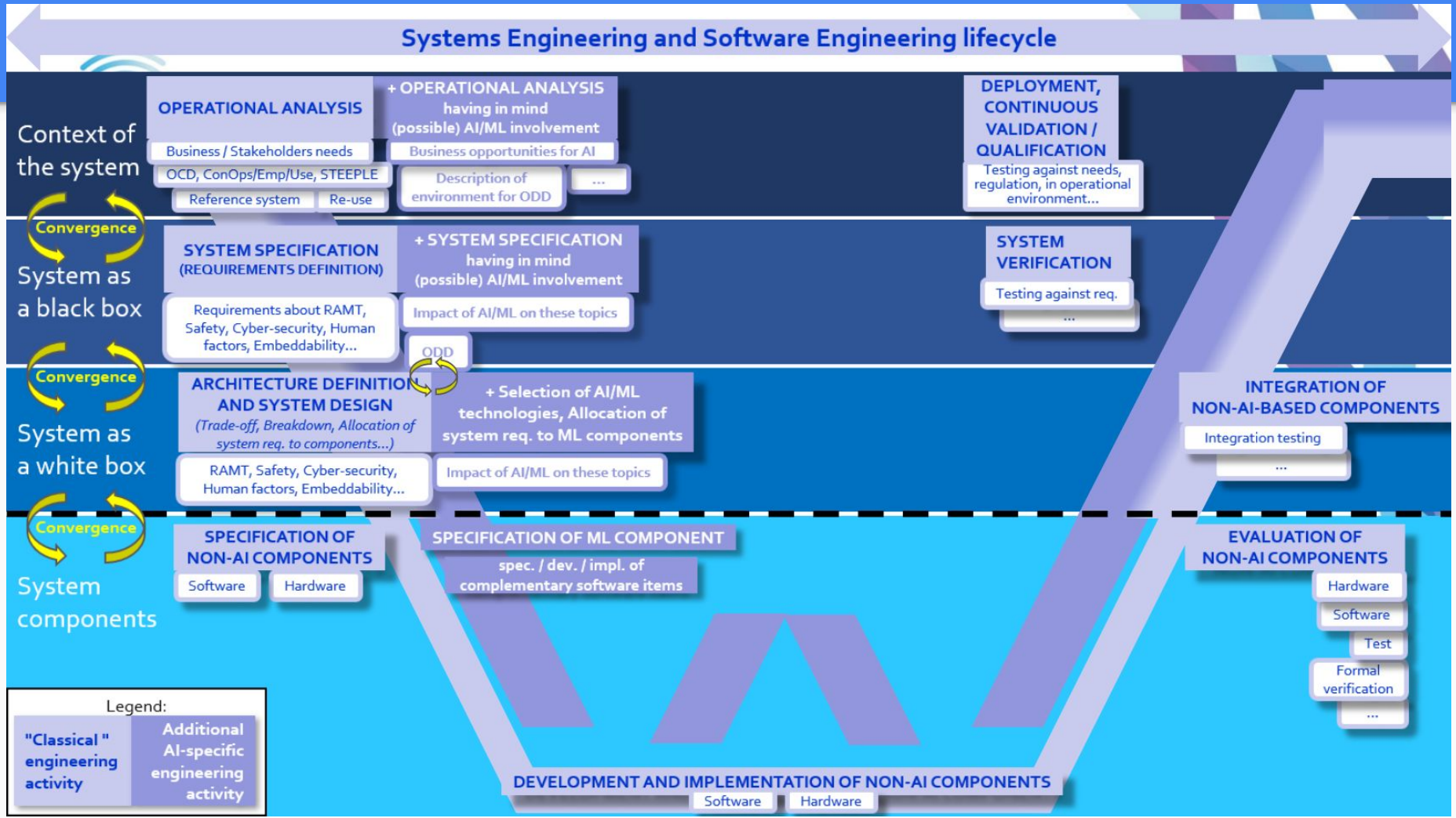

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+ OPERATIONAL ANALYSIS having in mind (possible) AI/ML involvement
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Description of environment for ODD
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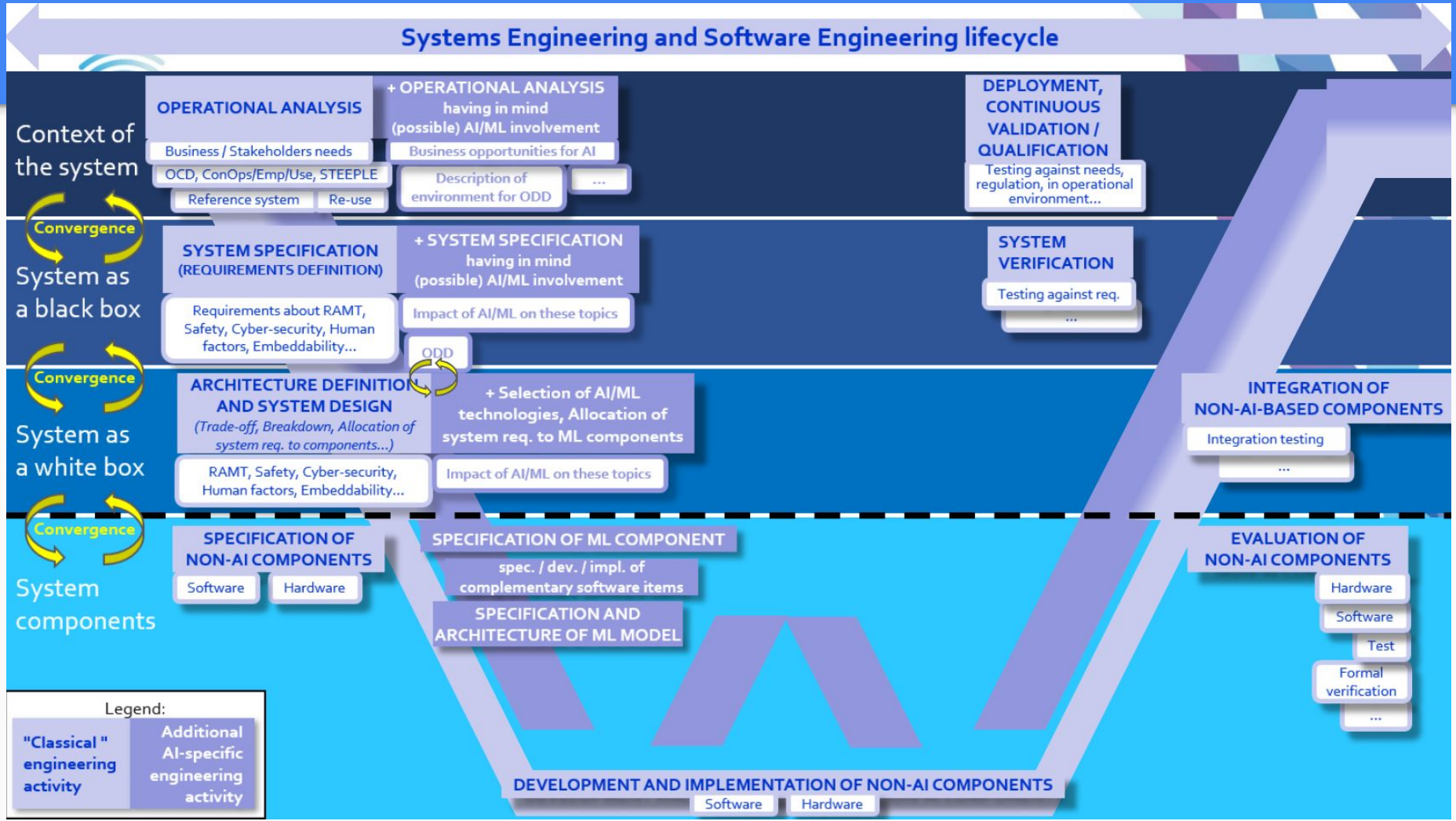

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Requirements about RAMT, Safety, Cyber-security, Human factors, Embeddability...
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Hardware
EVALUATION OF NON-AI COMPONENTS
Hardware
Software
Test
Formal verification
...
DEVELOPMENT AND IMPLEMENTATION OF NON-AI COMPONENTS
Software
Hardware
Legend:
"Classical " engineering activity
Additional AI-specific engineering activity

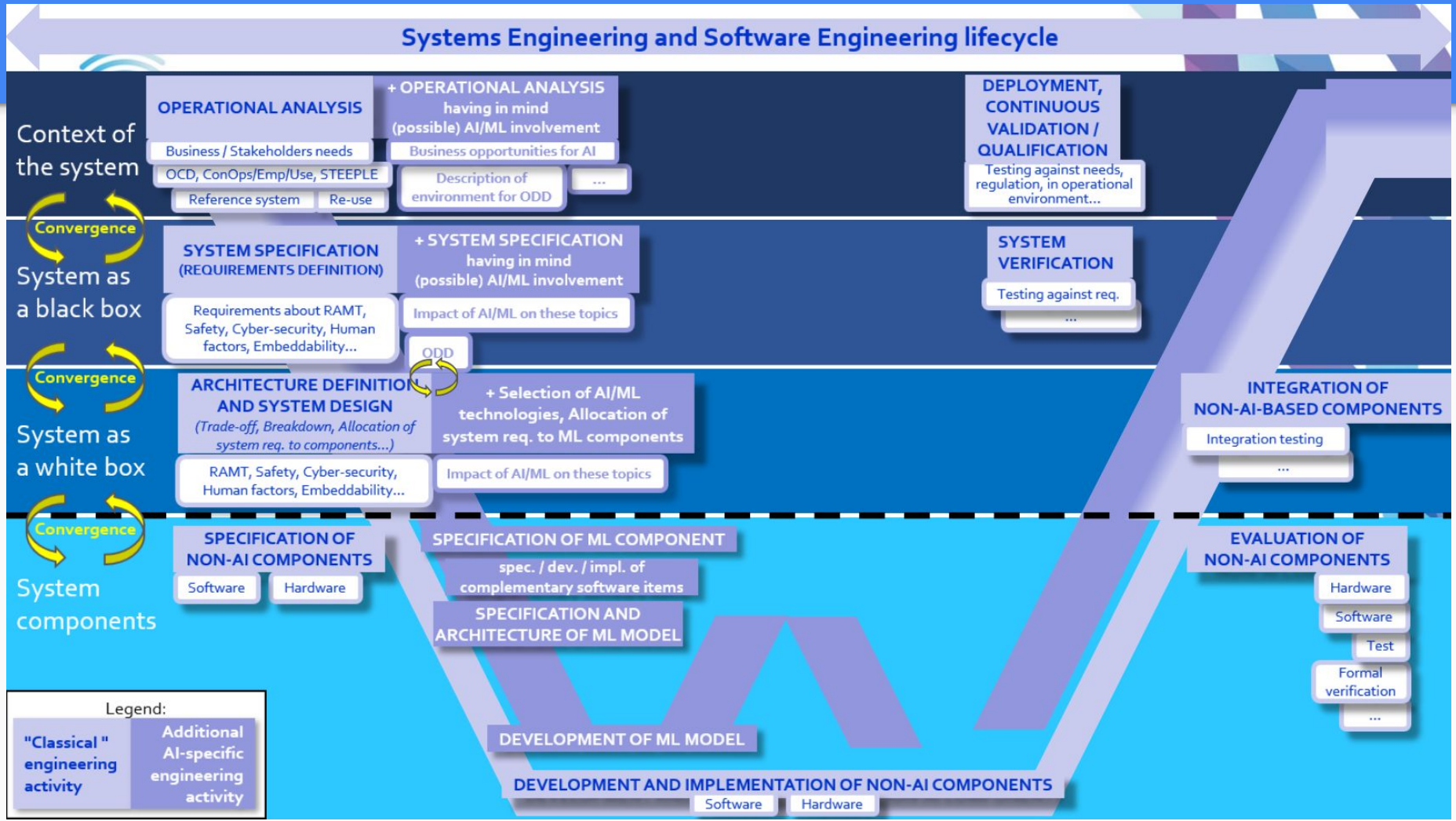

Systems Engineering and Software Engineering lifecycle
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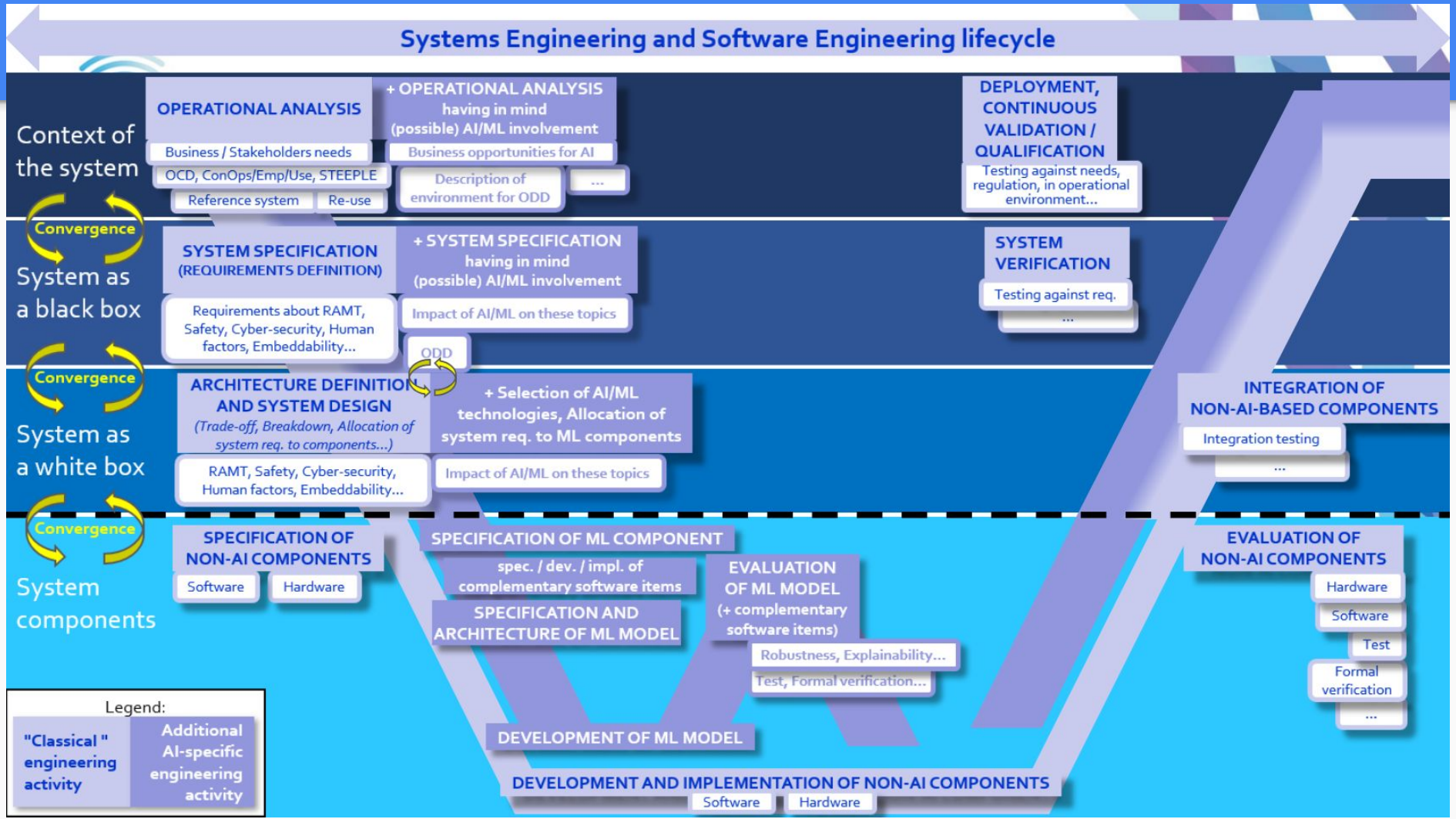

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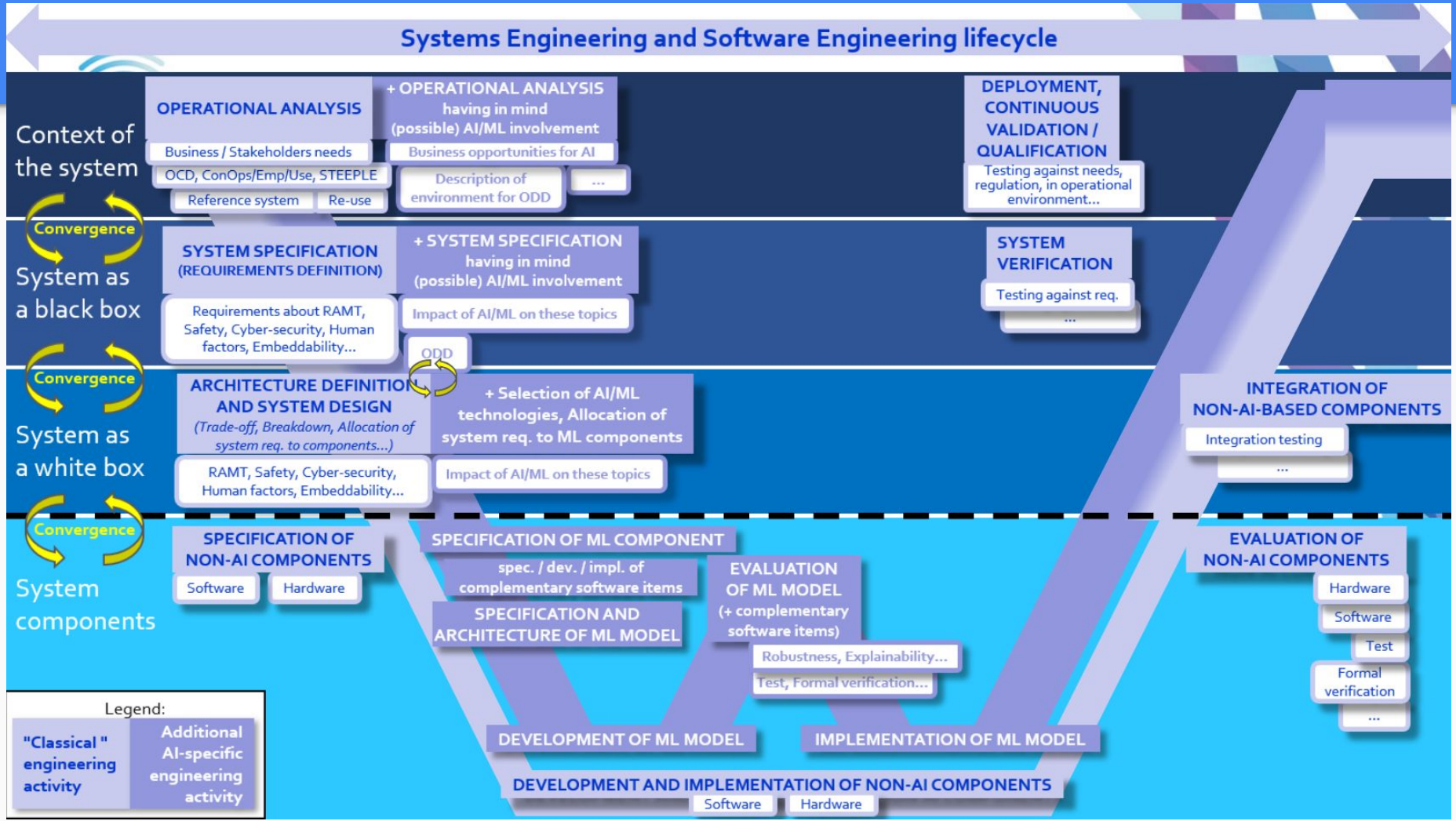

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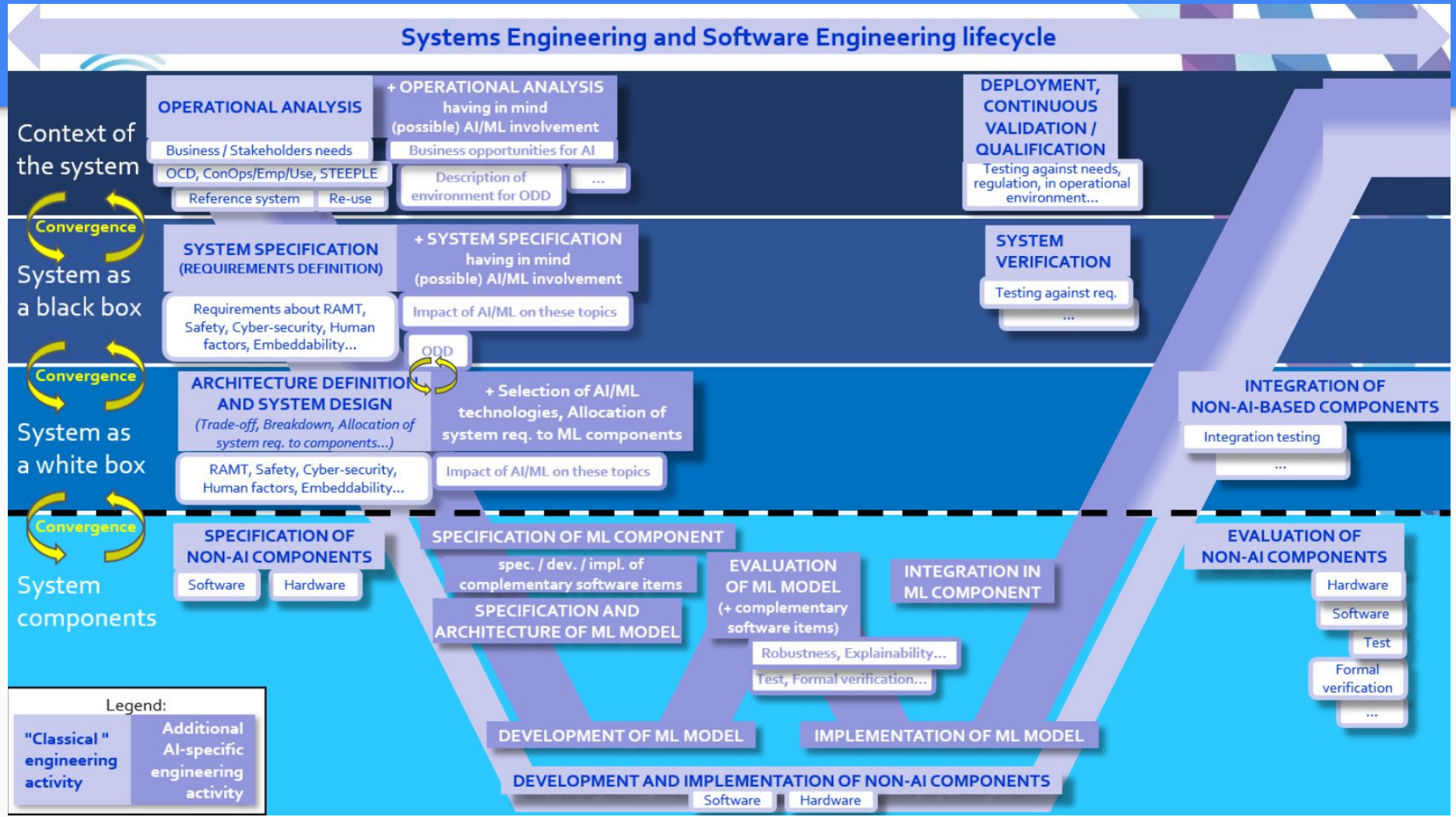

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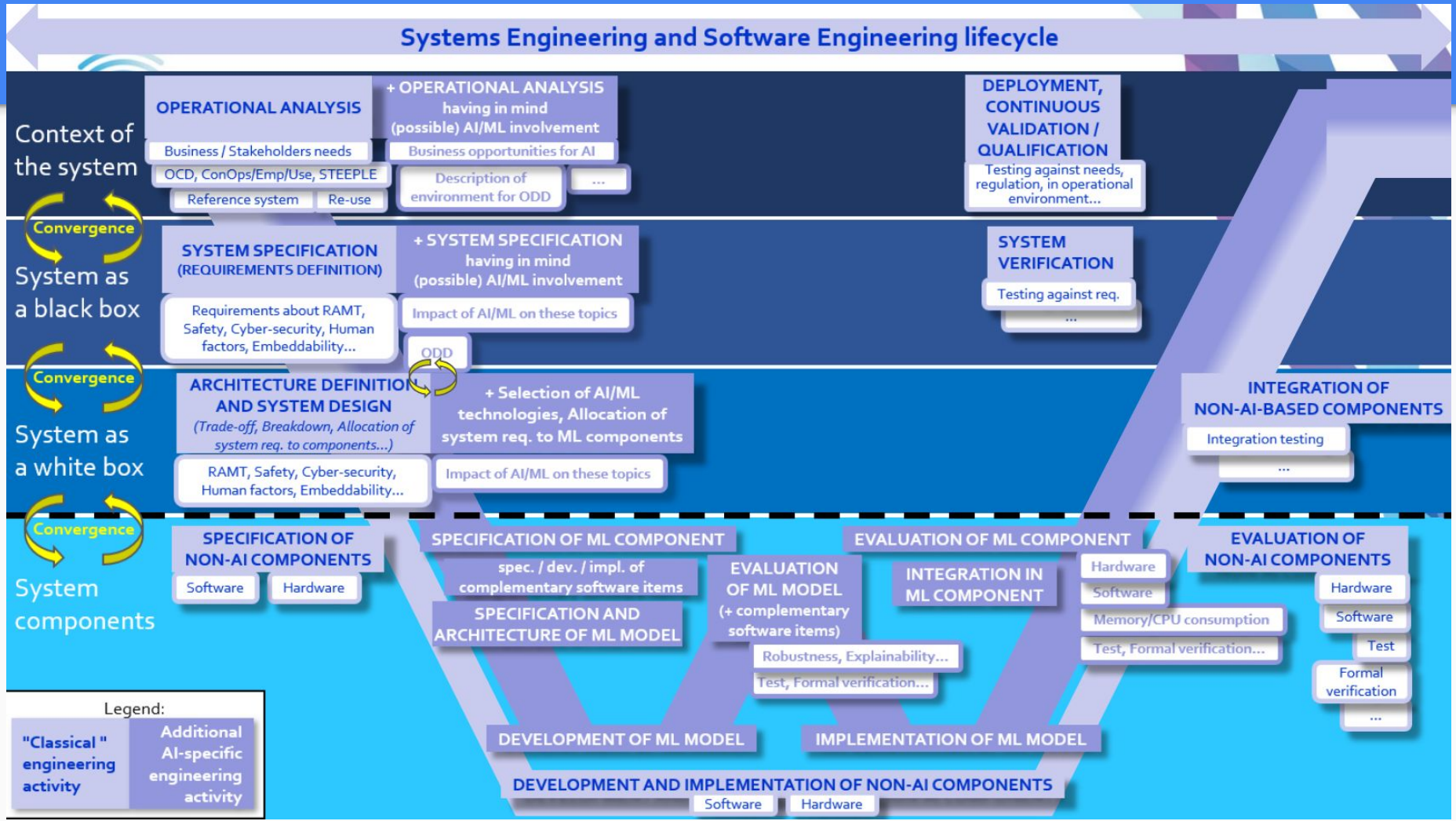

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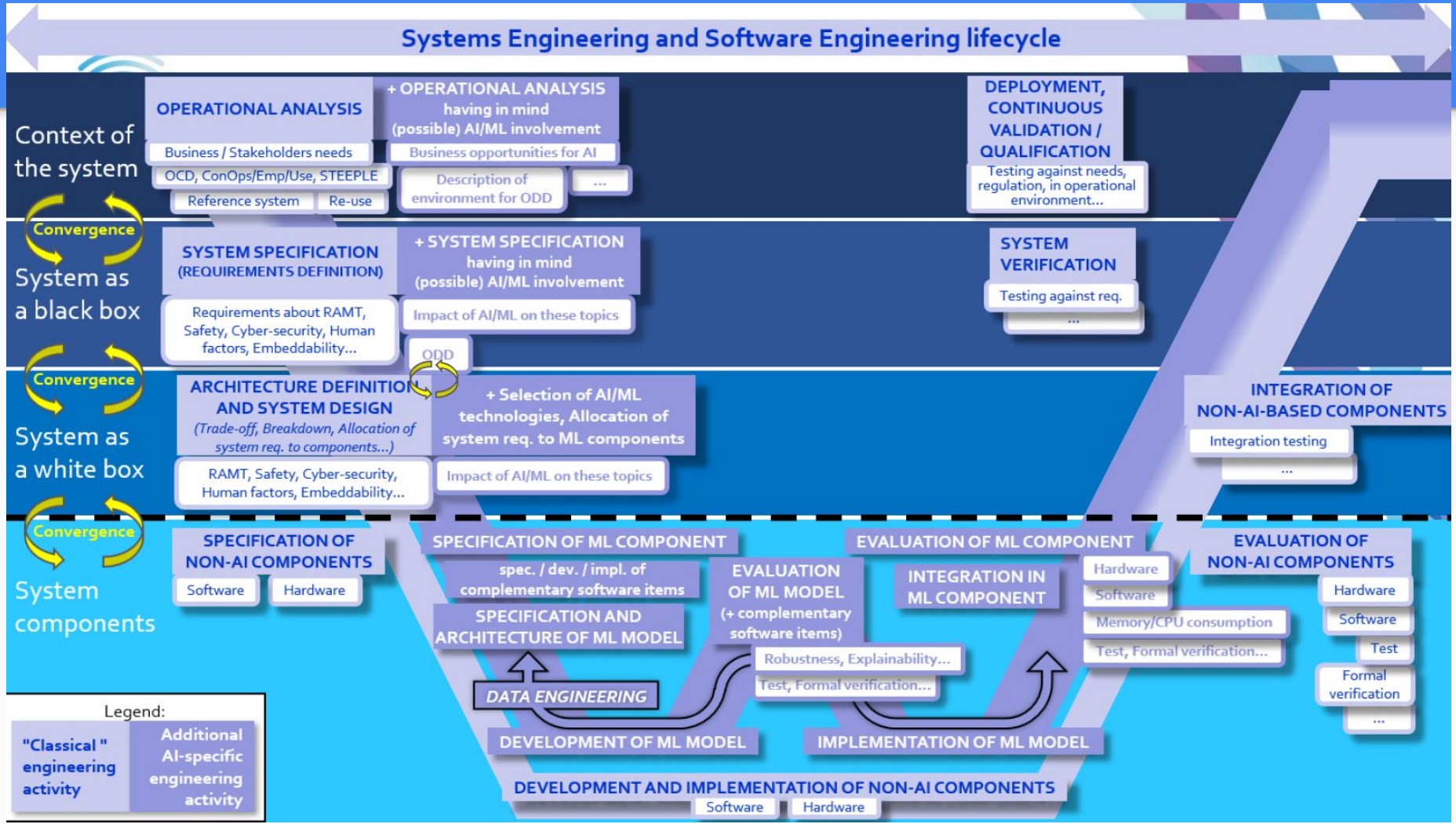

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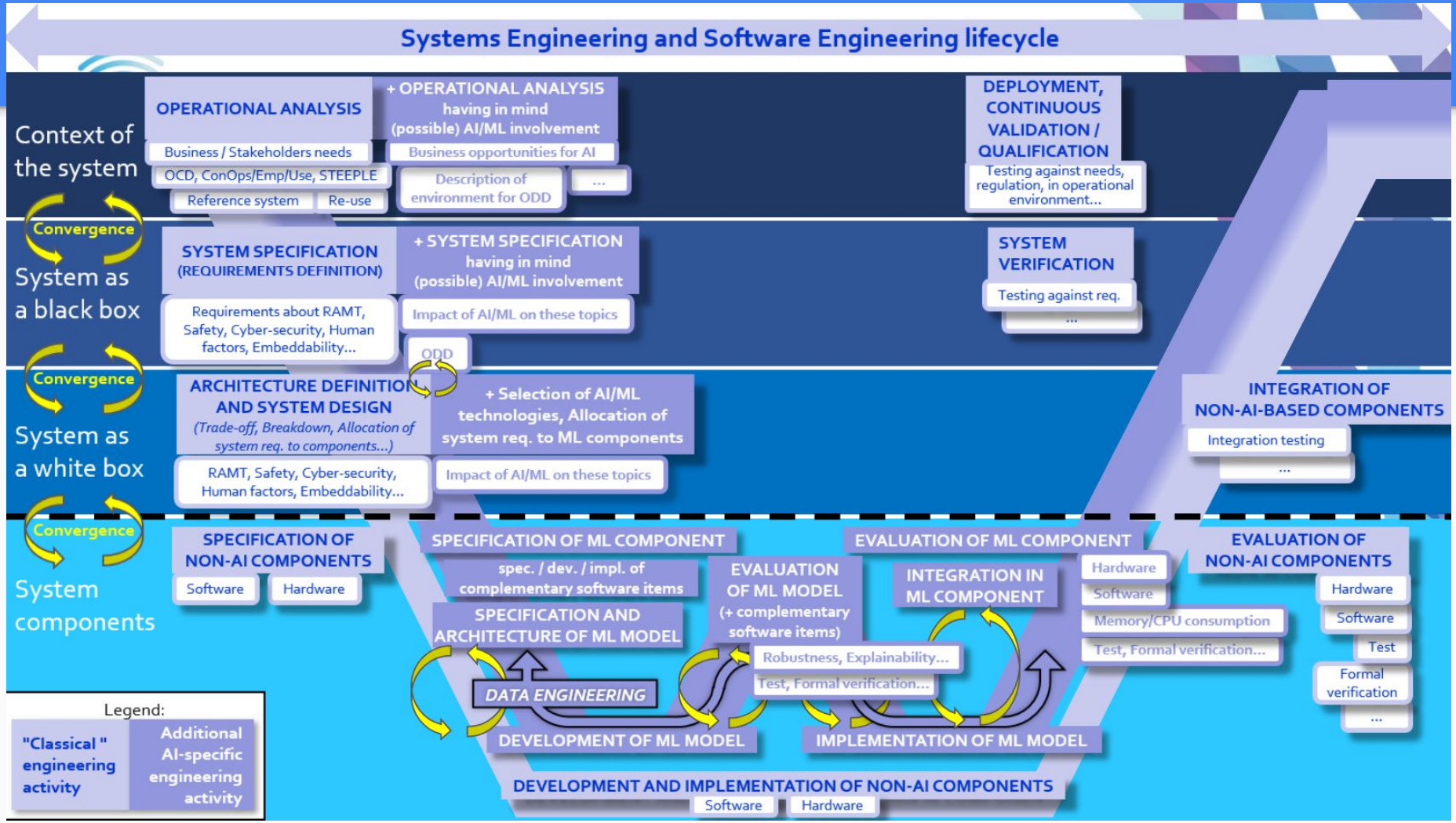

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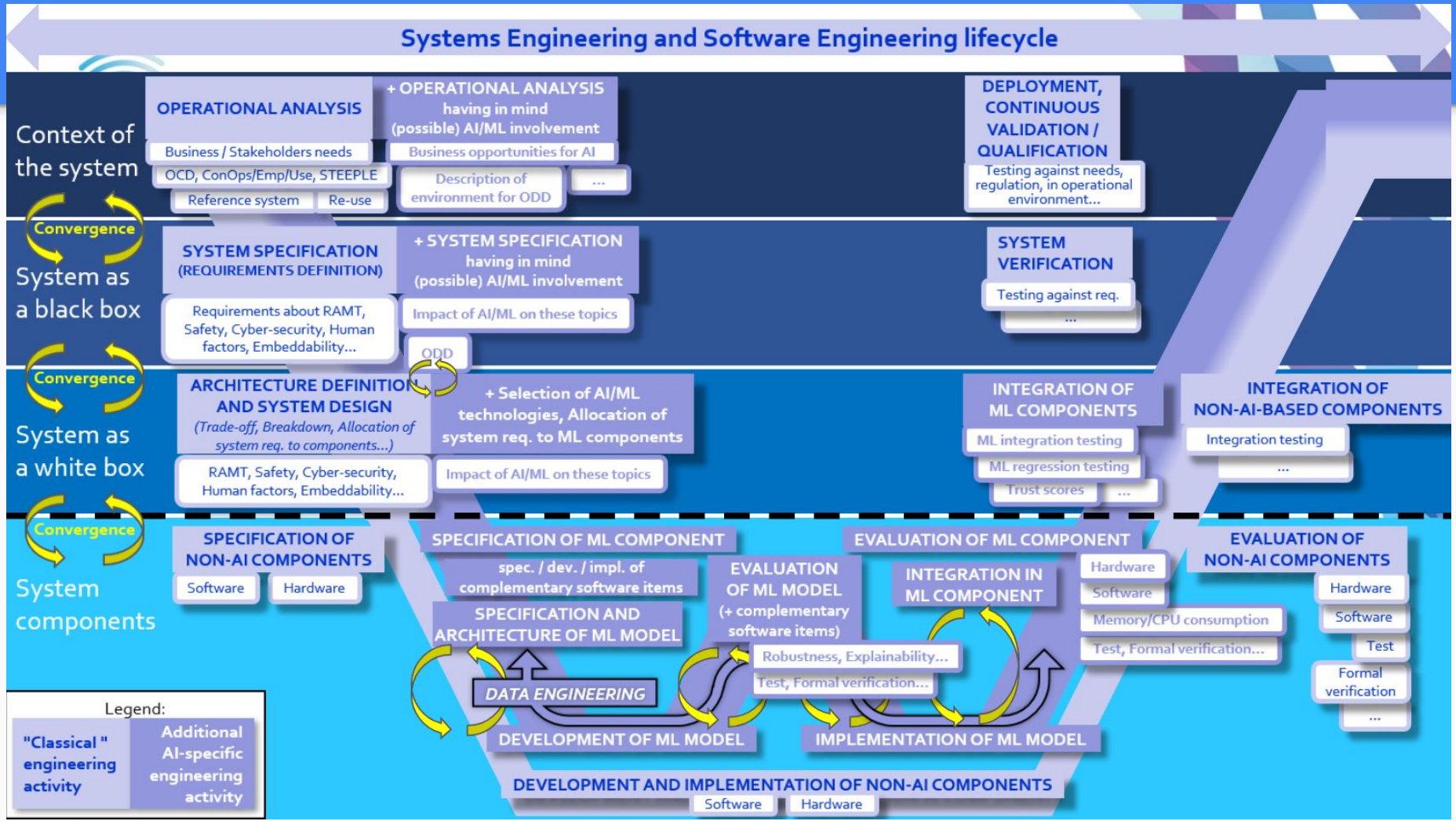

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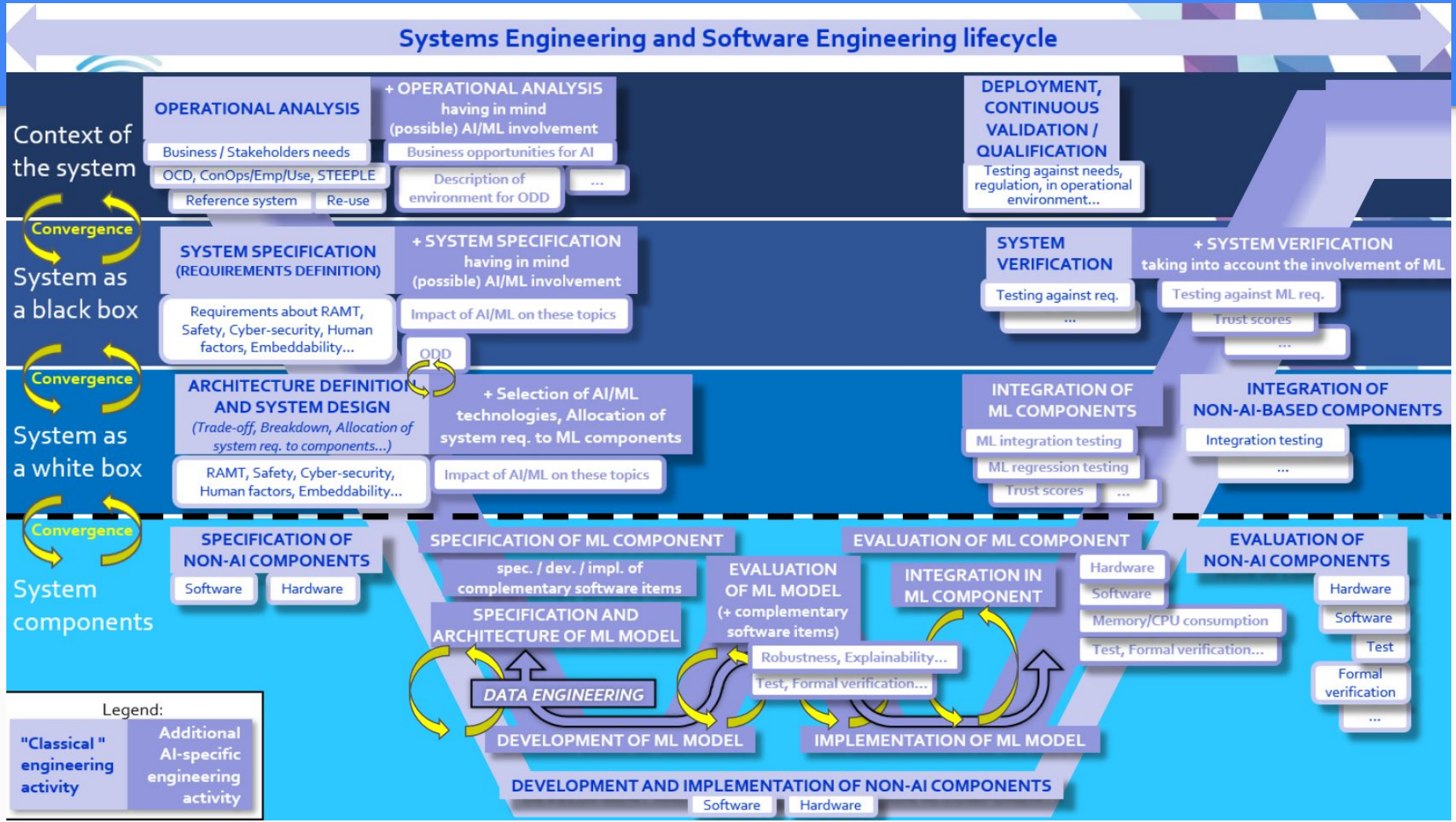

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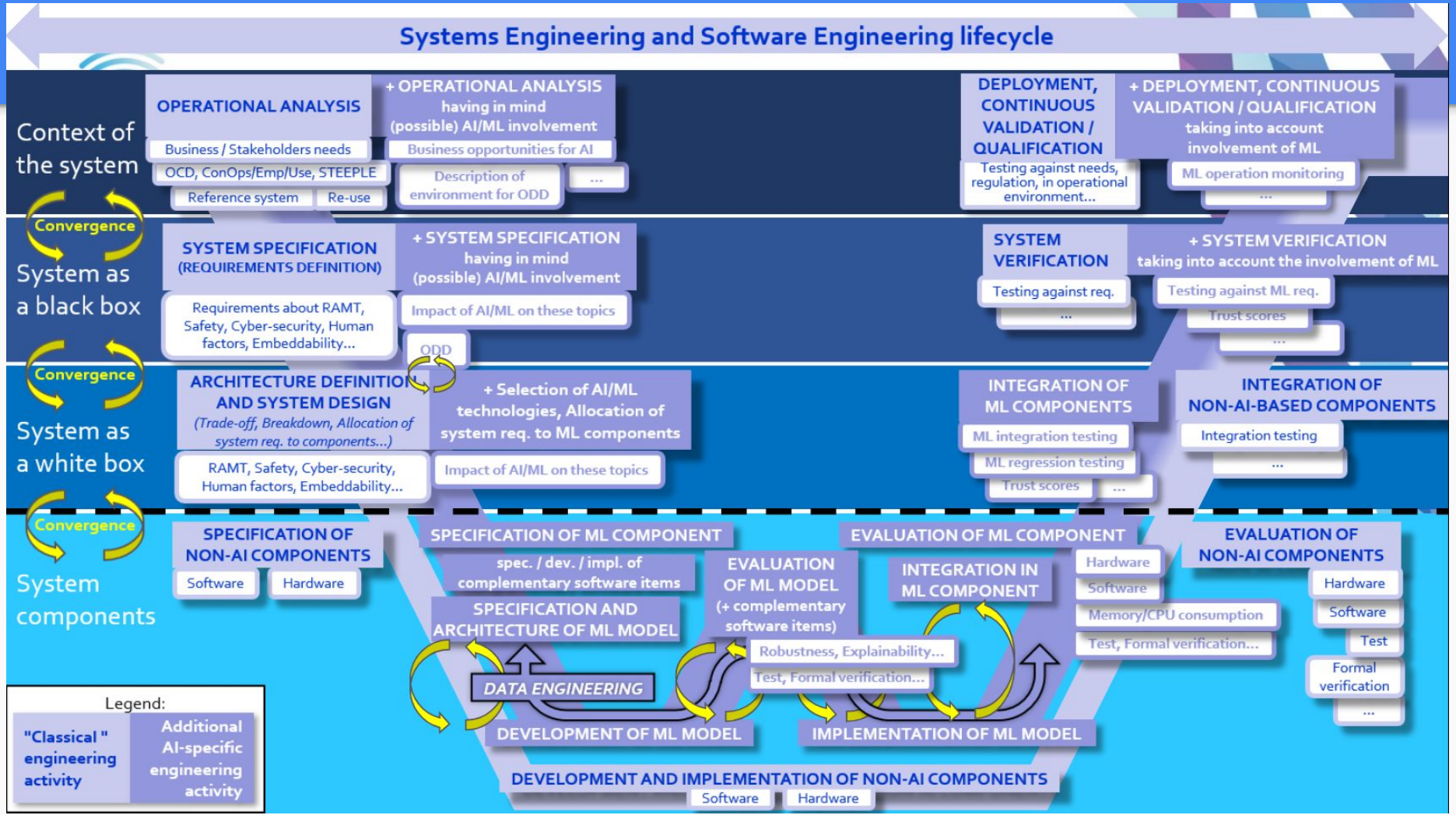

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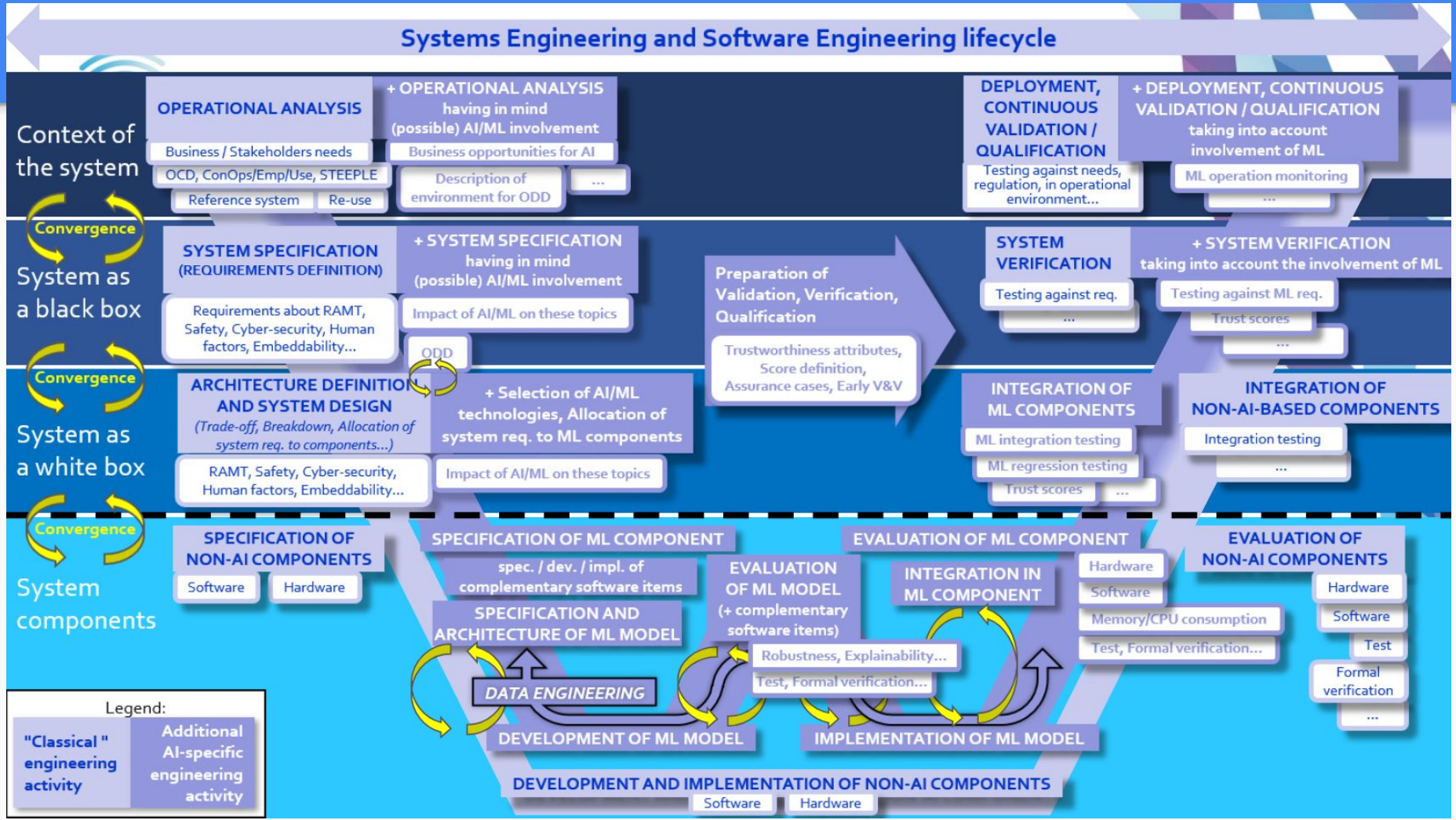

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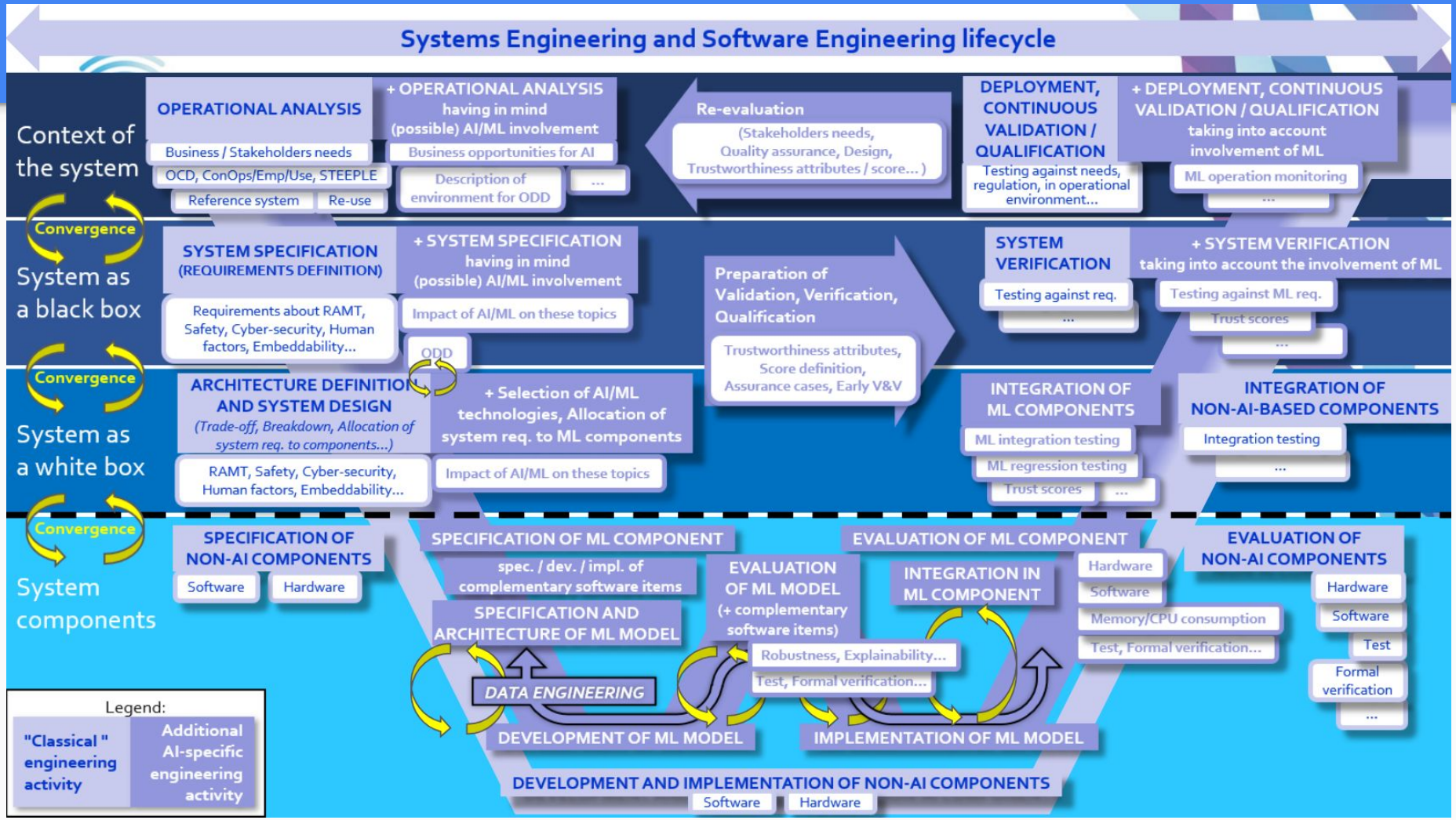

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Requirements about RAMT, Safety, Cyber-security, Human factors, Embeddability...
+ SYSTEM SPECIFICATION having in mind (possible) AI/ML involvement
Impact of AI/ML on these topics
ODD
SYSTEM VERIFICATION
Testing against req.
...
Convergence
System as a white box
ARCHITECTURE DEFINITION AND SYSTEM DESIGN
(Trade-off, Breakdown, Allocation of system req. to components...)
RAMT, Safety, Cyber-security, Human factors, Embeddability...
+ Selection of AI/ML technologies, Allocation of system req. to ML components
Impact of AI/ML on these topics
INTEGRATION OF ML COMPONENTS
ML integration testing
ML regression testing
Trust scores
...
INTEGRATION OF NON-AI-BASED COMPONENTS
Integration testing
...
Convergence
System components
SPECIFICATION OF NON-AI COMPONENTS
Software
Hardware
SPECIFICATION OF ML COMPONENT
spec. / dev. / impl. of complementary software items
SPECIFICATION AND ARCHITECTURE OF ML MODEL
DATA ENGINEERING
DEVELOPMENT OF ML MODEL
EVALUATION OF ML MODEL (+ complementary software items)
Robustness, Explainability...
Test, Formal verification...
EVALUATION OF ML COMPONENT
INTEGRATION IN ML COMPONENT
Hardware
Software
Memory/CPU consumption
Test, Formal verification...
IMPLEMENTATION OF ML MODEL
EVALUATION OF NON-AI COMPONENTS
Hardware
Software
Test
Formal verification
...
DEVELOPMENT AND IMPLEMENTATION OF NON-AI COMPONENTS
Software
Hardware
Legend:
"Classical" engineering activity
Additional AI-specific engineering activity

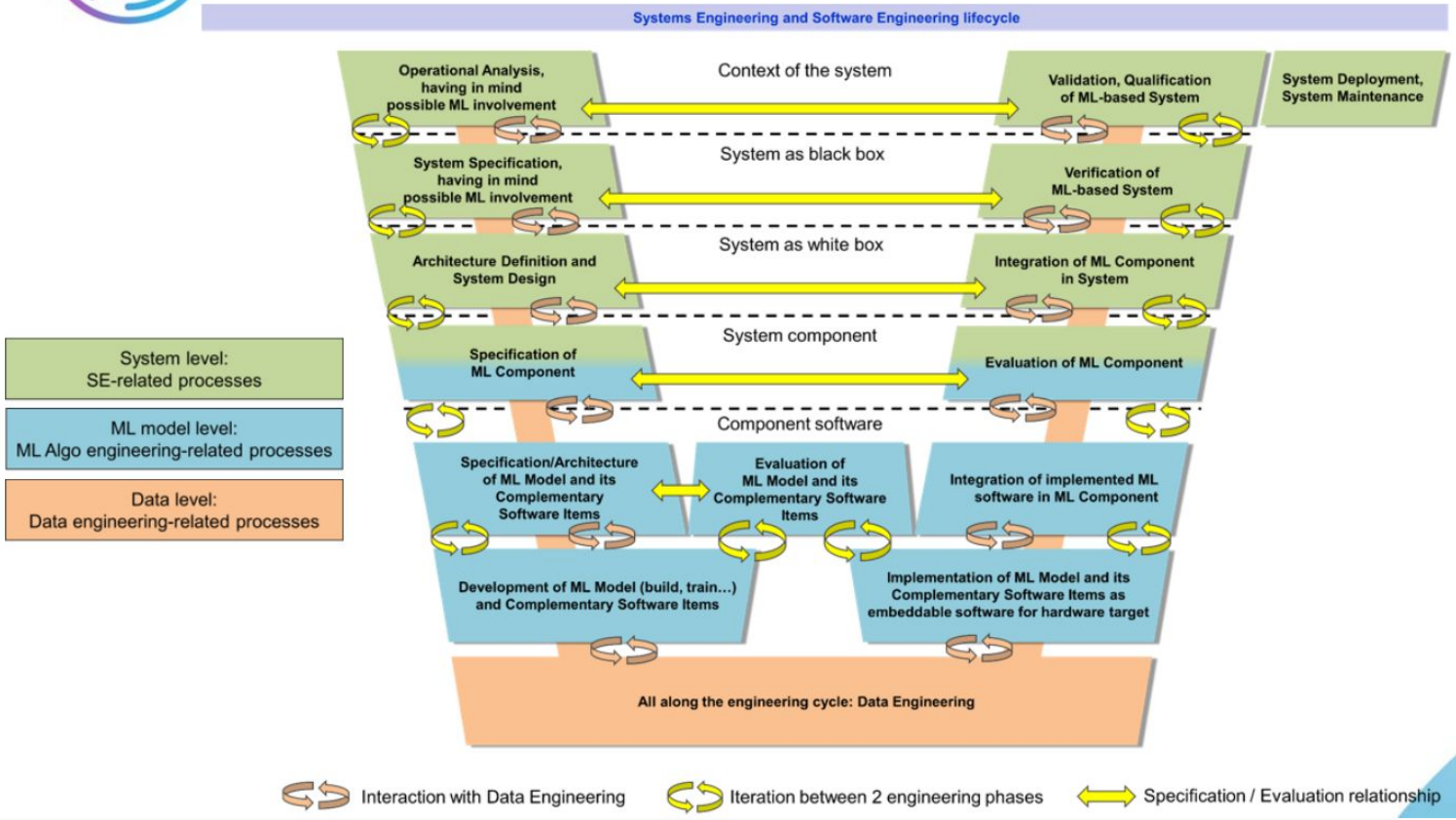

Systems Engineering and Software Engineering lifecycle
Context of the system
OPERATIONAL ANALYSIS
Business / Stakeholders needs
OCD, ConOps/Emp/Use, STEEPLE
Reference system
Re-use
+ OPERATIONAL ANALYSIS having in mind (possible) AI/ML involvement
Business opportunities for AI
Description of environment for ODD
...
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ML operation monitoring
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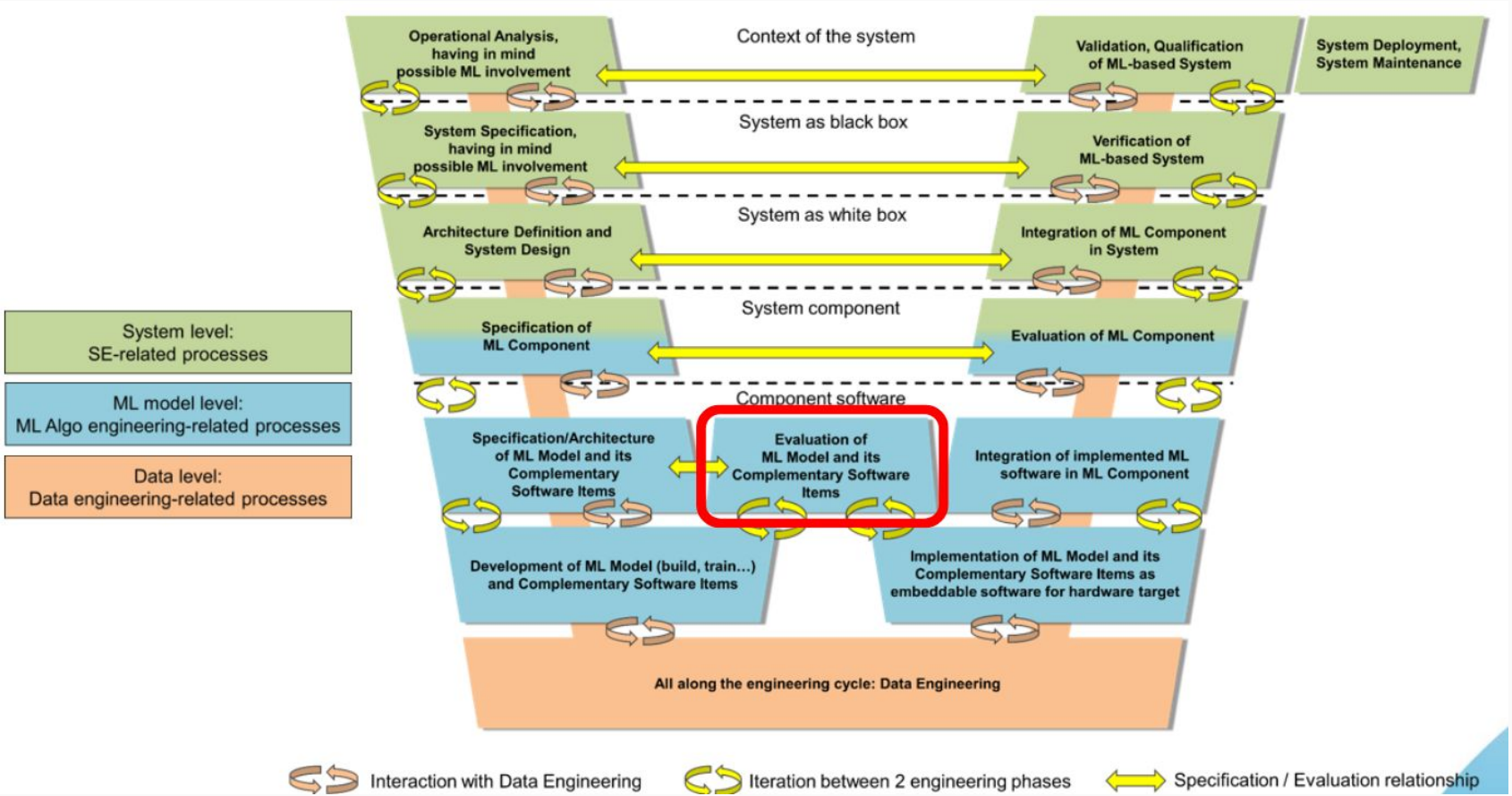

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Impact of AI/ML on these topics
ODD
Preparation of Validation, Verification, Qualification
Trustworthiness attributes, Score definition, Assurance cases, Early V&V
SYSTEM VERIFICATION
Testing against req.
...
+ SYSTEM VERIFICATION taking into account the involvement of ML
Testing against ML req.
Trust scores
...
Convergence
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...
Re-evaluation
(Stakeholders needs, Quality assurance, Design, Trustworthiness attributes / score...)
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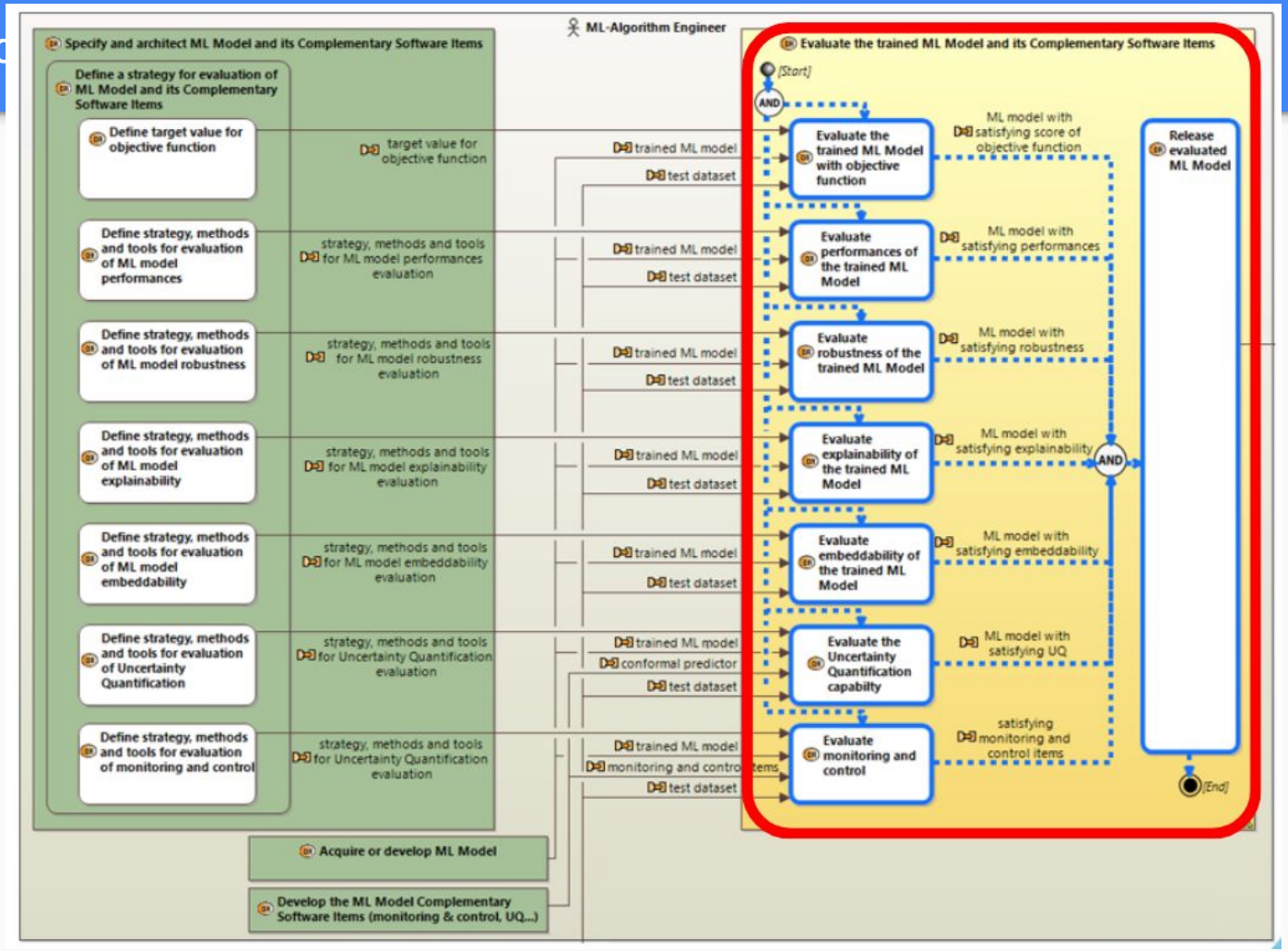
Highest view of the method

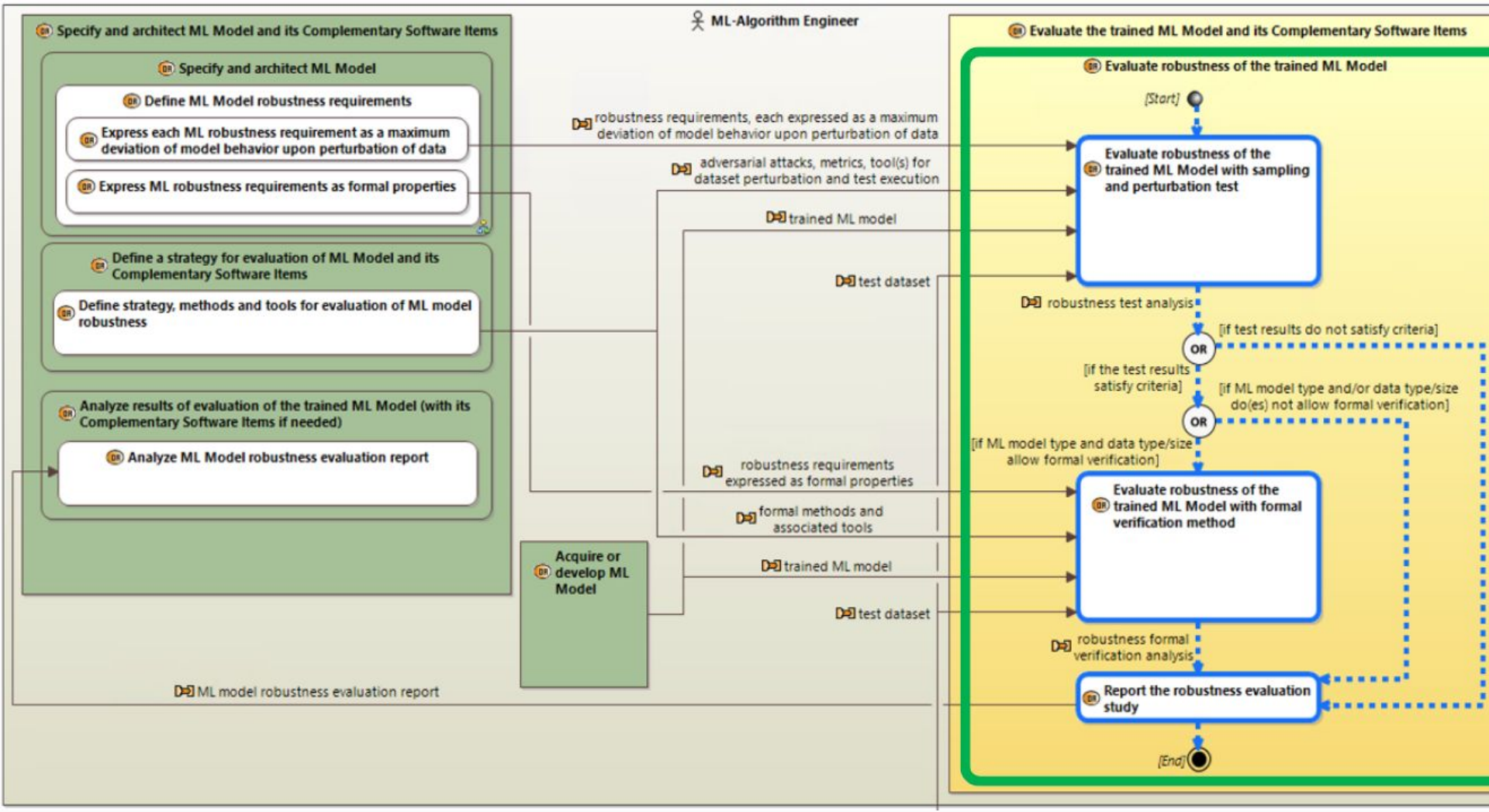


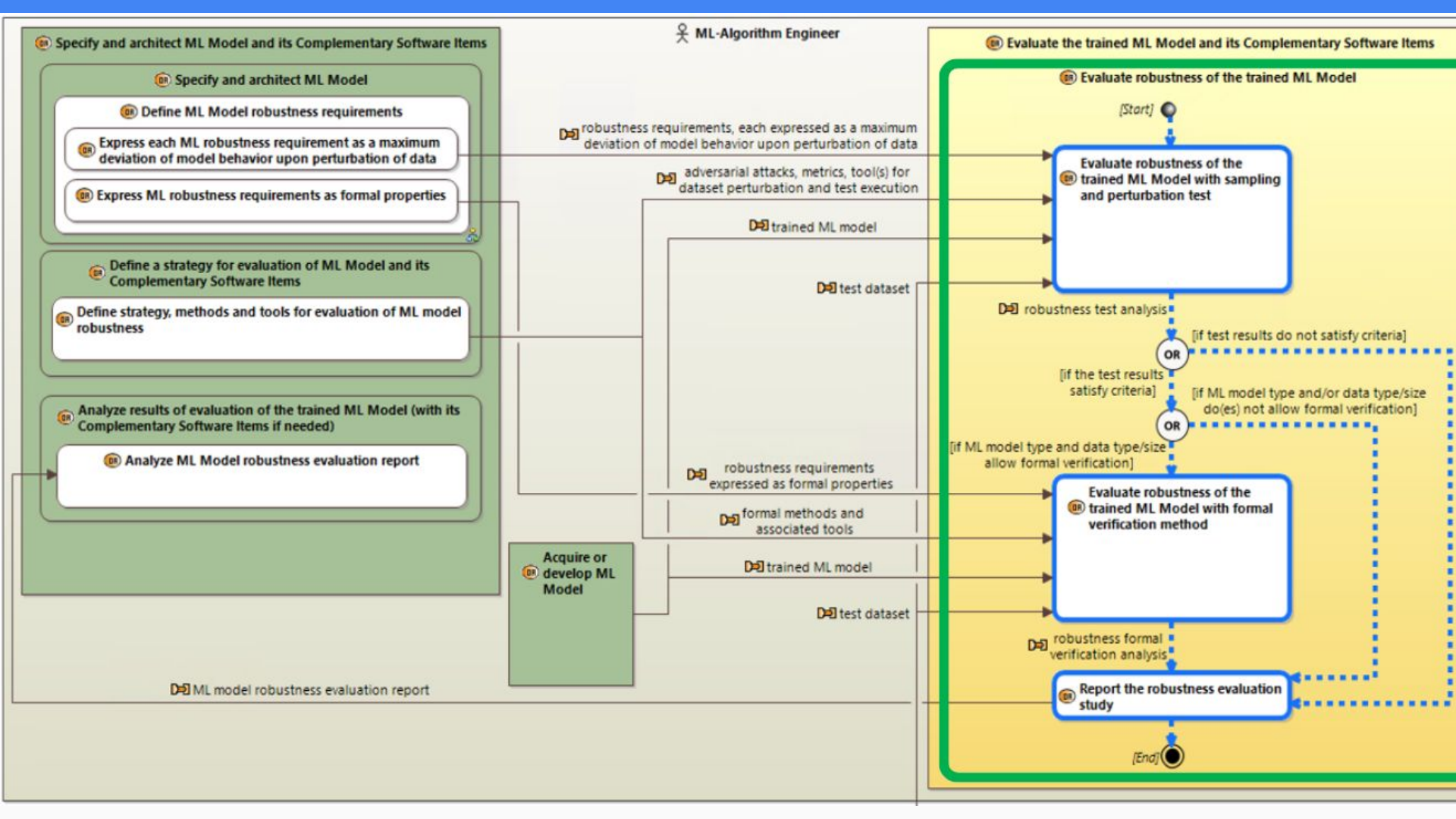
End-to-end: complexity

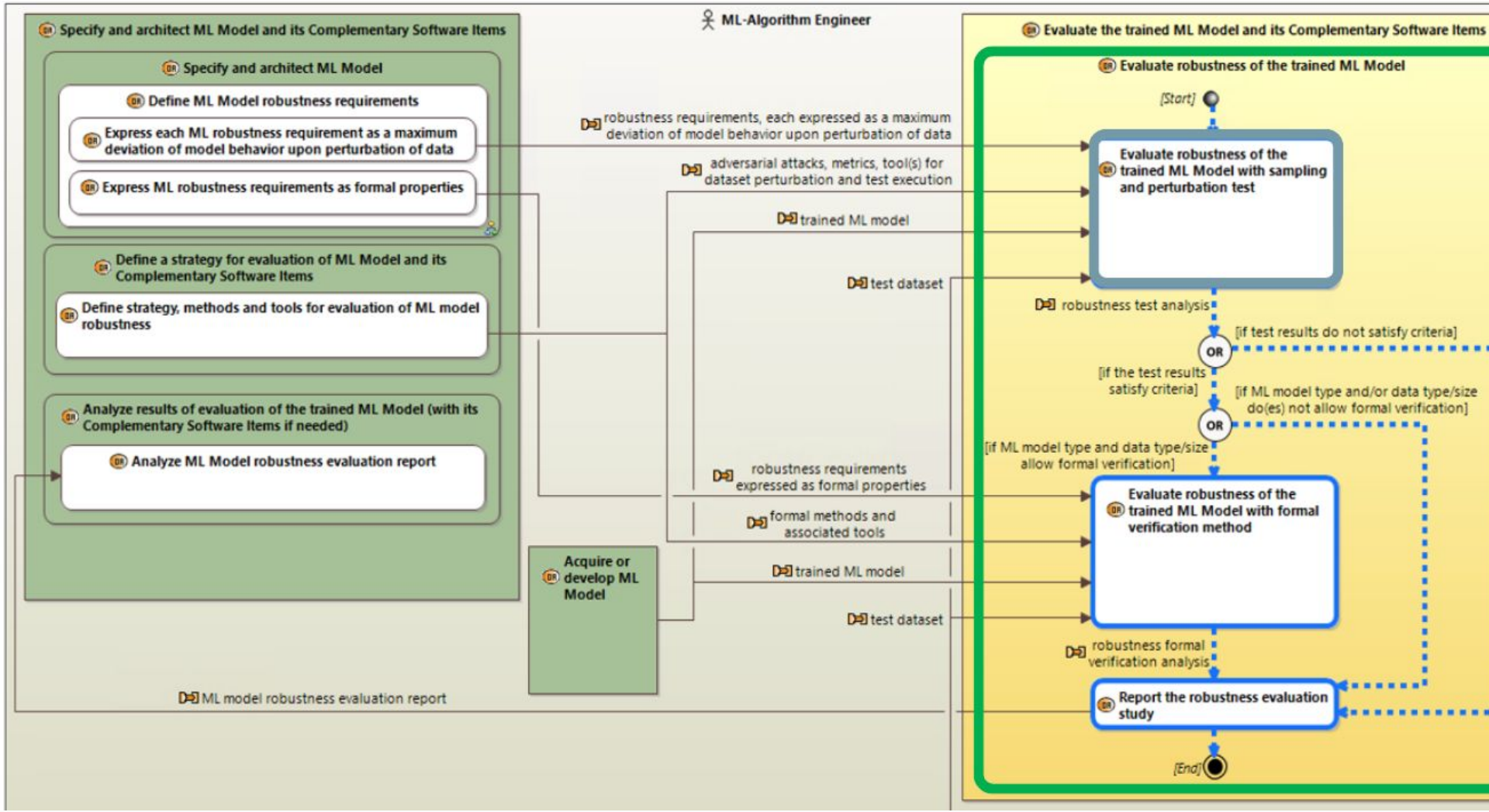


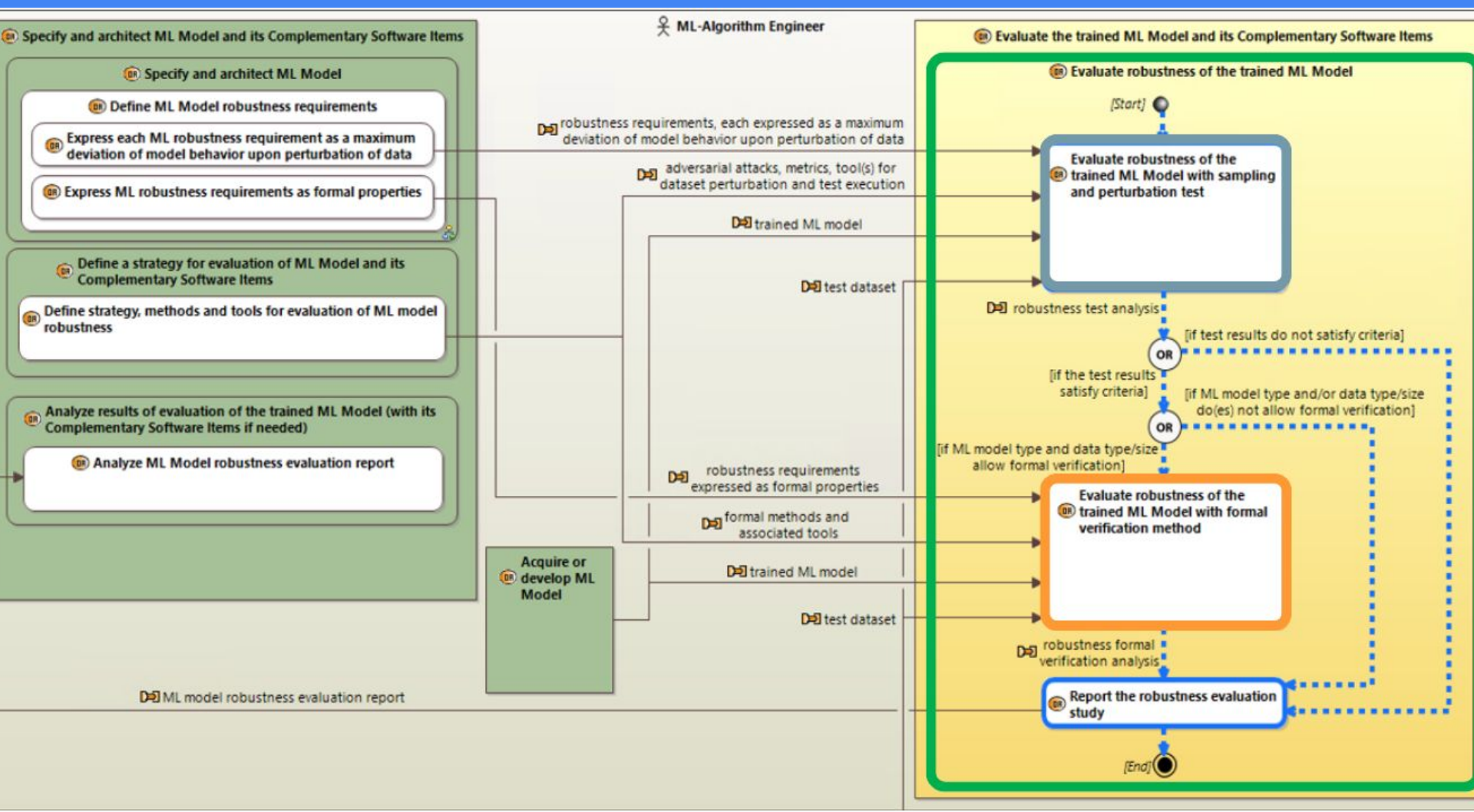
End-to-end


ML-Algorithm Engineer
Specify and architect ML Model and its Complementary Software Items
Define a strategy for evaluation of ML Model and its Complementary Software Items
Define target value for objective function
target value for objective function
Define strategy, methods and tools for evaluation of ML model performances
strategy, methods and tools for ML model performances evaluation
Define strategy, methods and tools for evaluation of ML model robustness
strategy, methods and tools for ML model robustness evaluation
Define strategy, methods and tools for evaluation of ML model explainability
strategy, methods and tools for ML model explainability evaluation
Define strategy, methods and tools for evaluation of ML model embeddability
strategy, methods and tools for ML model embeddability evaluation
Define strategy, methods and tools for evaluation of Uncertainty Quantification
strategy, methods and tools for Uncertainty Quantification evaluation
Define strategy, methods and tools for evaluation of monitoring and control
strategy, methods and tools for Uncertainty Quantification evaluation
trained ML model
test dataset
trained ML model
test dataset
trained ML model
test dataset
trained ML model
test dataset
trained ML model
test dataset
trained ML model
conformal predictor
test dataset
trained ML model
monitoring and control items
test dataset
Acquire or develop ML Model
Develop the ML Model Complementary Software Items (monitoring & control, UQ...)
Evaluate the trained ML Model and its Complementary Software Items
[Start]
AND
Evaluate the trained ML Model with objective function
ML model with satisfying score of objective function
Evaluate performances of the trained ML Model
ML model with satisfying performances
Evaluate robustness of the trained ML Model
ML model with satisfying robustness
Evaluate explainability of the trained ML Model
ML model with satisfying explainability
AND
Evaluate embeddability of the trained ML Model
ML model with satisfying embeddability
Evaluate the Uncertainty Quantification capability
ML model with satisfying UQ
Evaluate monitoring and control
satisfying monitoring and control items
Release evaluated ML Model
[End]

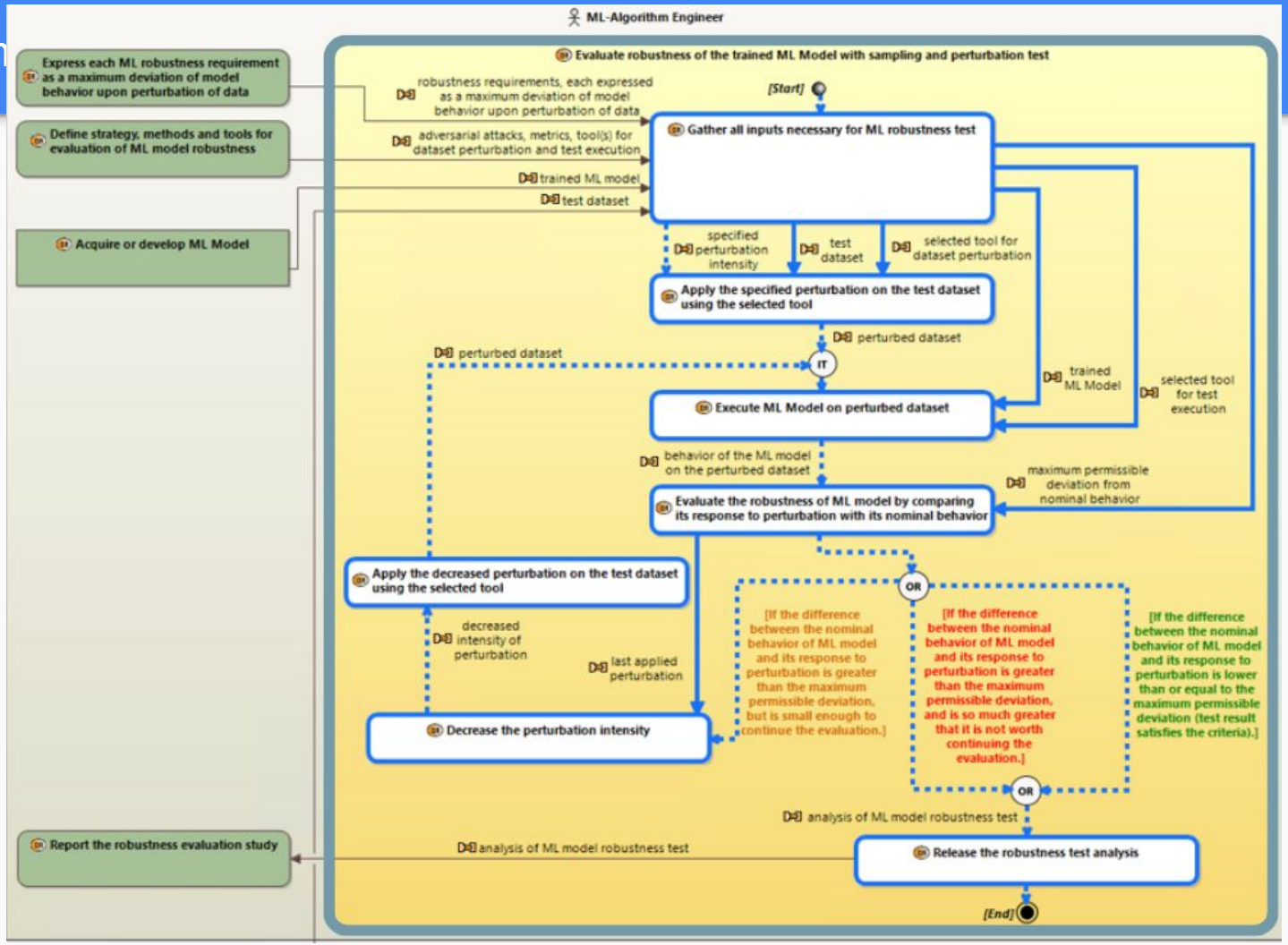

ML-Algorithm Engineer
Specify and architect ML Model and its Complementary Software Items
Specify and architect ML Model
Define ML Model robustness requirements
Express each ML robustness requirement as a maximum deviation of model behavior upon perturbation of data
Express ML robustness requirements as formal properties
Define a strategy for evaluation of ML Model and its Complementary Software Items
Define strategy, methods and tools for evaluation of ML model robustness
Analyze results of evaluation of the trained ML Model (with its Complementary Software Items if needed)
Analyze ML Model robustness evaluation report
Acquire or develop ML Model
robustness requirements, each expressed as a maximum deviation of model behavior upon perturbation of data
adversarial attacks, metrics, tool(s) for dataset perturbation and test execution
trained ML model
test dataset
robustness requirements expressed as formal properties
formal methods and associated tools
trained ML model
test dataset
ML model robustness evaluation report
Evaluate the trained ML Model and its Complementary Software Items
Evaluate robustness of the trained ML Model
[Start]
Evaluate robustness of the trained ML Model with sampling and perturbation test
robustness test analysis
[if test results do not satisfy criteria]
OR
[if the test results satisfy criteria]
[if ML model type and/or data type/size do(es) not allow formal verification]
OR
[if ML model type and data type/size allow formal verification]
Evaluate robustness of the trained ML Model with formal verification method
robustness formal verification analysis
Report the robustness evaluation study
[End]

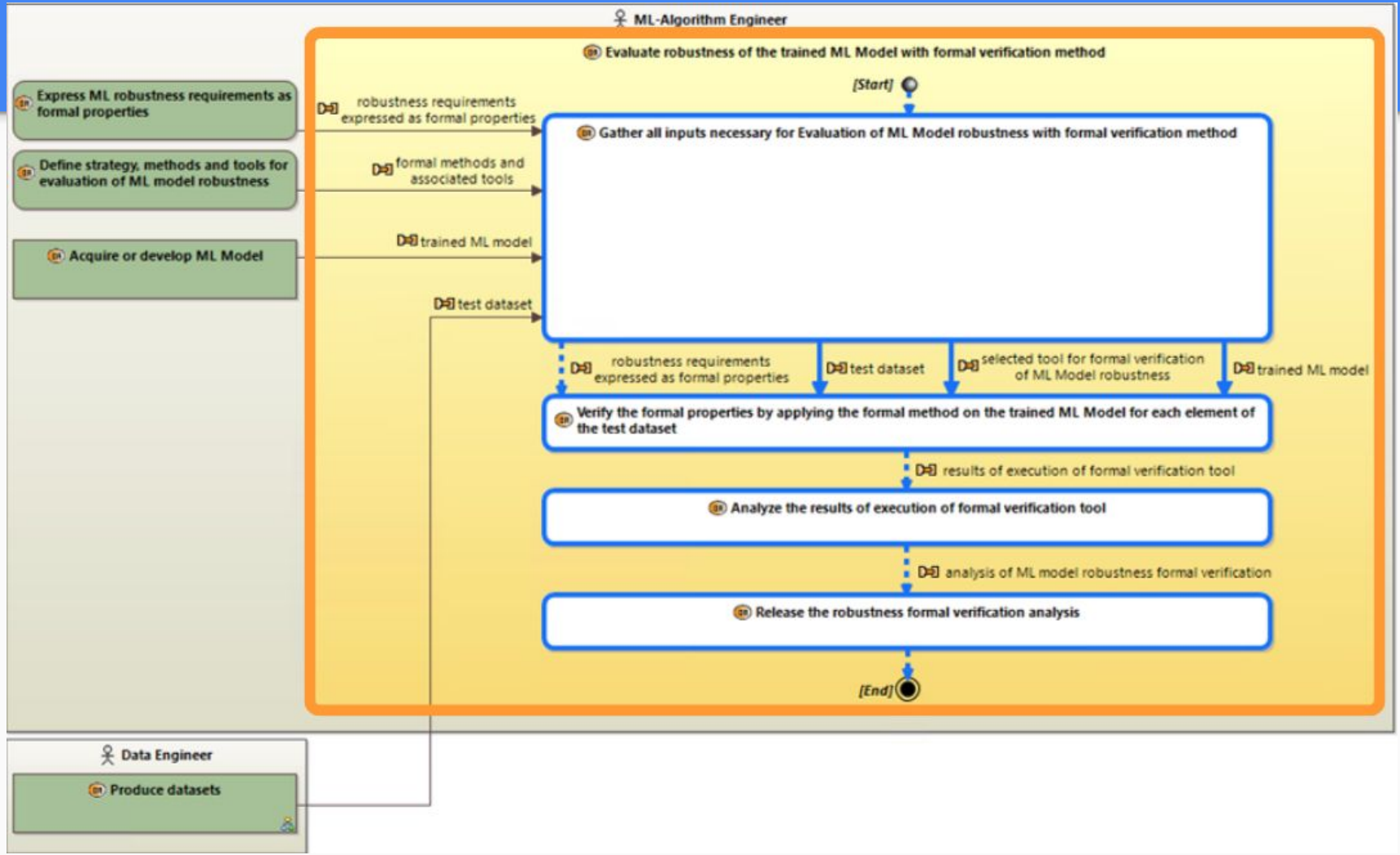

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Evaluate robustness of the trained ML Model with formal verification method
robustness formal verification analysis
Report the robustness evaluation study
[End]

End-to-en


ML-Algorithm Engineer
Express each ML robustness requirement as a maximum deviation of model behavior upon perturbation of data
Define strategy, methods and tools for evaluation of ML model robustness
Acquire or develop ML Model
Report the robustness evaluation study
Evaluate robustness of the trained ML Model with sampling and perturbation test
robustness requirements, each expressed as a maximum deviation of model behavior upon perturbation of data
adversarial attacks, metrics, tool(s) for dataset perturbation and test execution
trained ML model
test dataset
[Start]
Gather all inputs necessary for ML robustness test
specified perturbation intensity
test dataset
selected tool for dataset perturbation
Apply the specified perturbation on the test dataset using the selected tool
perturbed dataset
perturbed dataset
IT
trained ML Model
selected tool for test execution
Execute ML Model on perturbed dataset
behavior of the ML model on the perturbed dataset
maximum permissible deviation from nominal behavior
Evaluate the robustness of ML model by comparing its response to perturbation with its nominal behavior
Apply the decreased perturbation on the test dataset using the selected tool
OR
[If the difference between the nominal behavior of ML model and its response to perturbation is greater than the maximum permissible deviation, but is small enough to continue the evaluation.]
[If the difference between the nominal behavior of ML model and its response to perturbation is greater than the maximum permissible deviation, and is so much greater that it is not worth continuing the evaluation.]
[If the difference between the nominal behavior of ML model and its response to perturbation is lower than or equal to the maximum permissible deviation (test result satisfies the criteria).]
decreased intensity of perturbation
last applied perturbation
Decrease the perturbation intensity
OR
analysis of ML model robustness test
analysis of ML model robustness test
Release the robustness test analysis
[End]


ML-Algorithm Engineer
Evaluate robustness of the trained ML Model with formal verification method
[Start]
Express ML robustness requirements as formal properties
Define strategy, methods and tools for evaluation of ML model robustness
Acquire or develop ML Model
robustness requirements expressed as formal properties
formal methods and associated tools
trained ML model
test dataset
Gather all inputs necessary for Evaluation of ML Model robustness with formal verification method
robustness requirements expressed as formal properties
test dataset
selected tool for formal verification of ML Model robustness
trained ML model
Verify the formal properties by applying the formal method on the trained ML Model for each element of the test dataset
results of execution of formal verification tool
Analyze the results of execution of formal verification tool
analysis of ML model robustness formal verification
Release the robustness formal verification analysis
[End]
Data Engineer
Produce datasets

End-to-end

C.1 “Operational Analysis from ML perspective”

C.2 “System Specification” from ML perspective”

C.3 “Architecture Definition and System Designfrom ML perspective ”

C.4 “Specification and architecture of ML Component”

C.5 “Specification and architecture of ML Model and its Complementary Software Items”

C.6 “Software items complementary to the ML Model (specification, development, implementation)”

C.7 “Data Engineering”

C.16 “System Continuous Re-evaluation from ML perspective”

C.15 “System Validation/Qualification from ML perspective”

C.14 “System Verification from ML perspective”

C.13 “Integration of ML Components in System”

C.12 “Evaluation of ML Component”

C.11 “Integration of implemented Software in ML Component”

C.10 “Implementation of ML Model as embeddable software”

C.9 “Evaluation of ML Model and its Complementary Software Items”

C.8 “Development of ML Model”

End-to-end

C.1 Analyse opérationnelle : Comprendre les besoins et les objectifs de l'entreprise.

C.2 Spécification du système : Définir les exigences fonctionnelles et non fonctionnelles du système d'IA.

C.3 Définition de l'architecture : Concevoir la structure et les composants du système d'IA.

C.4 Spécification et architecture du composant ML : Définir les exigences et la conception du composant ML.

C.5 Spécification et architecture du modèle ML : Définir les exigences et la conception du modèle ML.

C.6 Éléments logiciels complémentaires au modèle ML : Spécifier, développer et implémenter les logiciels nécessaires au support du modèle ML.

C.7 Ingénierie des données : Préparer, nettoyer et transformer les données pour l'entraînement et l'utilisation du modèle ML.

C.14 Vérification du système : Garantir que le système d'IA est construit conformément aux spécifications.

C.15 Validation/Qualification du système : Garantir que le système d'IA répond aux besoins et aux objectifs de l'entreprise.

C.16 Réévaluation continue du système : Surveiller et évaluer en permanence les performances du système d'IA et l'adapter aux

Atelier

Dans quelques jours va se tenir le COMEX qui validera la roadmap produit de votre entreprise.
Vous allez cadrer le projet afin de convaincre les dirigeants que votre produit possède un fort potentiel.

Ministère de la sécurité image

Appel à projet

Classification d’images d’une caméra d’une Raspberry

Objectifs:

Détecter les personnes présentes dans des images fournies
En live capturé par un micro au Raspberry Pi.
Classifier la catégorie.

Atelier

Fonctionnalités attendues:

Détection en temps réel des sons.

Affichage du nombre de sons différents détectés.

Possibilité de sauvegarder les sons détectés.

2. Application du cycle en V:

Phase descendante:

Analyse des besoins:

Identifier les besoins des utilisateurs et les fonctionnalités à implémenter.

Définir les exigences de performance et de sécurité.

Identifier les contraintes matérielles et logicielles du Raspberry Pi.

Livrable: Document d'analyse des besoins.

Spécifications:

Rédiger les spécifications fonctionnelles et techniques du projet.

Définir l'architecture du système et les interfaces entre les différents modules.

Décrire les algorithmes de Computer Vision à utiliser.

Livrable: Cahier des charges.

Conception:

Concevoir l'interface utilisateur et l'architecture logicielle du projet.

Définir la structure des données et les classes à utiliser.

Développer les prototypes des modules principaux.

Livrable: Spécifications de conception.

Atelier

Développement:

Implémenter le code source du projet en utilisant un langage de programmation adapté (Python, C++).

Tester et déboguer le code pour corriger les erreurs.

Livrable: Code source du projet.

Atelier

Phase ascendante:

Tests unitaires:

Tester chaque module du projet individuellement pour vérifier son fonctionnement.

Livrable: Rapport de tests unitaires.

Tests d'intégration:

Tester l'interaction entre les différents modules du projet.

Livrable: Rapport de tests d'intégration.

Tests de validation:

Tester le système complet avec des cas d'utilisation réels.

Valider la conformité du système aux exigences du projet.

Livrable: Rapport de tests de validation.

Recette:

Déployer le système sur le Raspberry Pi et effectuer des tests en conditions réelles.

Livrable: Système finalisé et documenté

Juridique

RGPD

Le Règlement Général sur la Protection des Données (RGPD) est une réglementation européenne qui vise à protéger les données personnelles des citoyens européens. Il s'applique à toute organisation qui traite des données personnelles, y compris les organisations qui développent et utilisent des systèmes d'IA.

Points clés du RGPD pour la gestion de projet IA:

Collecte et traitement des données: Les données personnelles doivent être collectées et traitées de manière licite, loyale et transparente.

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Points clés du RGPD pour la gestion de projet IA:

Collecte et traitement des données: Les données personnelles doivent être collectées et traitées de manière licite, loyale et transparente.

Consentement: Les personnes doivent consentir au traitement de leurs données personnelles.

Minimisation des données: Les organisations ne doivent collecter que les données personnelles qui sont nécessaires à la réalisation de la finalité pour laquelle elles sont collectées.

Sécurité des données: Les organisations doivent mettre en place des mesures techniques et organisationnelles pour protéger les données personnelles contre la perte, le vol et l'abus.

Droit d'accès, de rectification et d'effacement: Les personnes ont le droit d'accéder à leurs données personnelles, de les rectifier et de les faire effacer.

Propriété intellectuelle

La propriété intellectuelle (PI) est un domaine du droit qui protège les créations de l'esprit humain. Cela inclut les inventions, les dessins industriels, les marques, les droits d'auteur et les secrets commerciaux.

Points clés de la PI pour la gestion de projet IA:

- Brevets: Les inventions peuvent être protégées par des brevets.
- Droits d'auteur: Les logiciels et les autres œuvres littéraires et artistiques sont protégés par des droits d'auteur.
- Secrets commerciaux: Les informations confidentielles qui procurent un avantage concurrentiel peuvent être protégées en tant que secrets

AI Act

L'IA Act est une proposition de réglementation européenne qui vise à réguler le développement et l'utilisation de l'IA. L'IA Act est toujours en cours de négociation, mais il devrait avoir un impact important sur la gestion de projet IA.

Points clés de l'IA Act pour la gestion de projet IA:

- **Classification des systèmes d'IA**: L'IA Act classe les systèmes d'IA en fonction de leur niveau de risque.
- **Exigences pour les systèmes d'IA**: Les systèmes d'IA à haut risque devront répondre à des exigences spécifiques en matière de sécurité, de transparence et d'éthique.
- **Gouvernance de l'IA**: L'IA Act crée un nouveau cadre de gouvernance pour l'IA.